UNIT - 1

Introduction to NDT & Radiographic Testing

1. Introduction to Non-Destructive Testing (NDT)

* **NDT** = examining a material, component or weld for defects / properties ***without damaging it or impairing its future usefulness*** .
* Also called **NDE** (Non-Destructive Evaluation) or **NDI** (Non-Destructive Inspection).
* The part is tested and then goes back into service - unlike tensile, impact or hardness tests which destroy / damage the sample.
* Used for: detecting flaws, measuring thickness, finding leaks, checking material properties, quality control.

Destructive vs Non-Destructive Testing

Point	Destructive Test	Non-Destructive Test
Sample	Damaged or broken	Remains usable
Testing done on	Sample / specimen from lot	100% of components possible
Cost	High (loss of material)	Low to moderate
Examples	Tensile, impact, fatigue, bend	RT, UT, MPT, LPT, ECT
Information	Actual strength values	Presence, size, position of flaws
Repeatability	Cannot repeat on same piece	Can be repeated many times

Objectives / Need of NDT

* Ensure ***safety and reliability*** of the product (boilers, aircraft, bridges, pipelines).
* Detect defects in early stage and ***reduce cost of rejection*** .
* Prevent accidents and failure in service.
* Maintain uniform quality, support process control.
* Helps ***life extension*** of old plants by in-service inspection.
* Required by codes and standards (ASME, API, IS, ISO).

Types of Defects

* **Surface defects** - cracks, laps, seams, pits, undercut, spatter.
* **Sub-surface defects** - just below the surface (porosity, inclusion).
* **Internal / volumetric defects** - blowholes, shrinkage cavity, slag, voids.
* **Planar defects** are more dangerous than volumetric ones (sharp, stress raiser).

Common NDT Methods

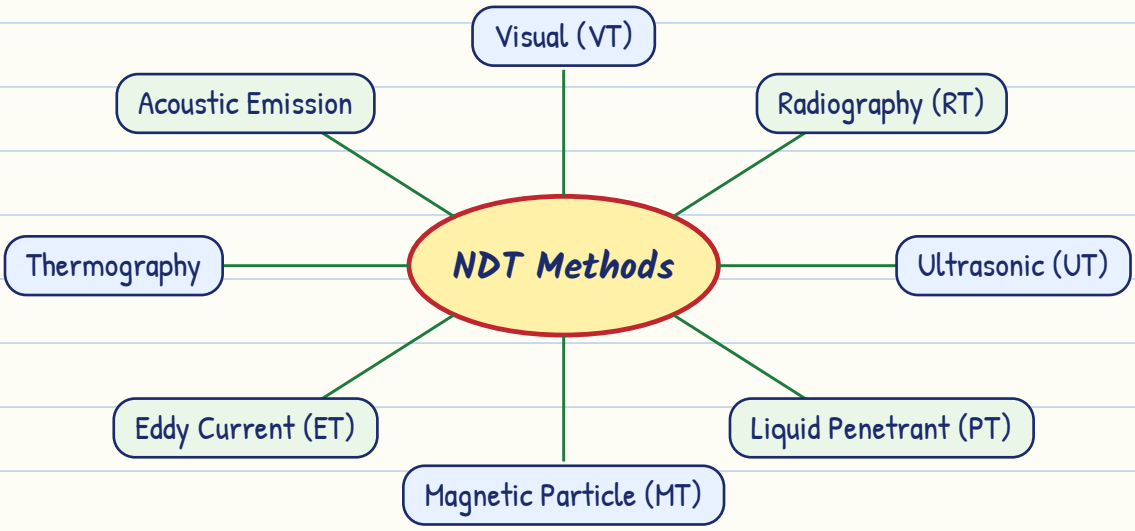


Fig: Principal NDT methods

* **Surface methods:** VT, PT, MT (and ET for near-surface).
* **Volumetric methods:** RT and UT detect internal flaws.

> **Remember:**
> Every NDT method has a **limited range of flaws it can detect** - so often two methods are combined (e.g. UT + MT, RT + PT) for complete inspection.

2. Radiographic Testing (RT) - Principle

* RT uses **short wavelength electromagnetic radiation** (X-rays or Gamma rays) that can penetrate opaque materials.
* Radiation passes through the specimen; **different parts absorb different amounts** .
* Thin section / void / crack absorbs **less** -> more radiation reaches film -> **darker** image.
* Thick section / dense inclusion absorbs **more** -> **lighter** image on film.

* The image recorded on film (radiograph) is a **shadow picture** of internal structure.
* Intensity after absorption: **I = I0 e^(-mu x)** where mu = linear attenuation coefficient, x = thickness, I0 = incident intensity.

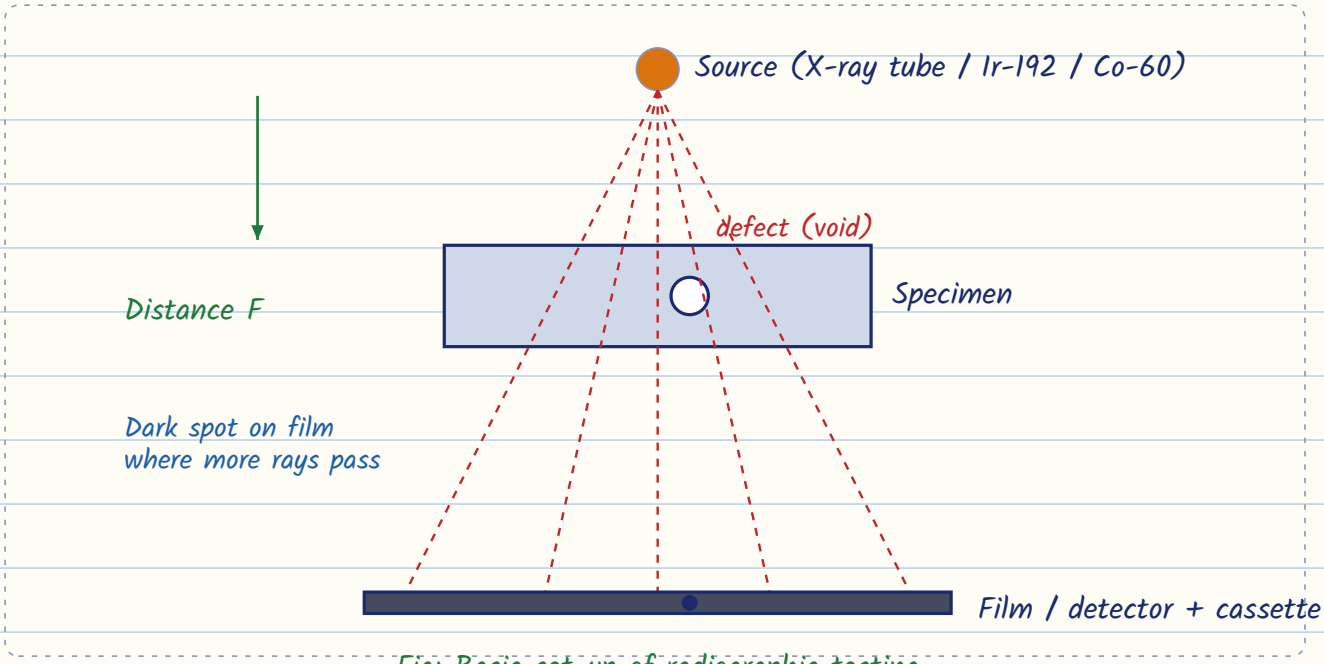


Fig: Basic set-up of radiographic testing

Steps in a Radiographic Test

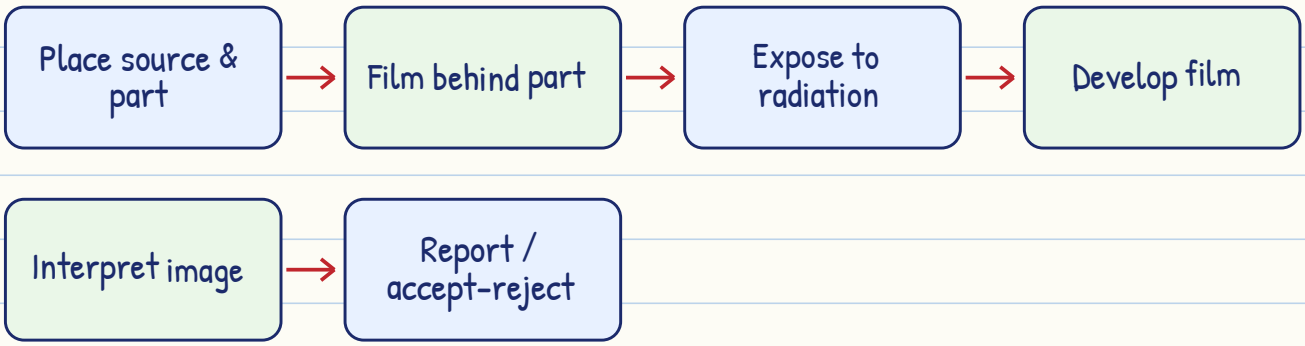


Fig: Radiographic procedure

* Film is loaded in a light-tight **cassette** with lead screens, placed on the side opposite to the source.
* **Penetrameter (IQI)** is kept on the source side to prove the sensitivity of the radiograph.

3. Electromagnetic Radiation – Basics

* X-rays and gamma rays are part of the electromagnetic spectrum; wavelength about **0.001 nm to 10 nm** .

* Travel with speed of light, in straight lines, cannot be deflected by electric or magnetic field.
* They are **invisible, ionising** , affect photographic film and cause fluorescence.
* Energy of a photon: $E = h f = h c / \lambda$ (h = Planck constant).
* Shorter wavelength = higher energy = more penetrating power.
* Quantity measured in keV or MeV; **hard X-rays** (short wavelength) penetrate more, **soft X-rays** penetrate less.

4. Sources of X-rays

4.1 Production of X-rays

* X-rays are produced when **high-speed electrons are suddenly stopped** by a heavy metal target.
* About **99% of energy becomes heat** and only about 1% becomes X-rays - hence cooling is a must.

4.2 X-ray Tube (Coolidge tube)

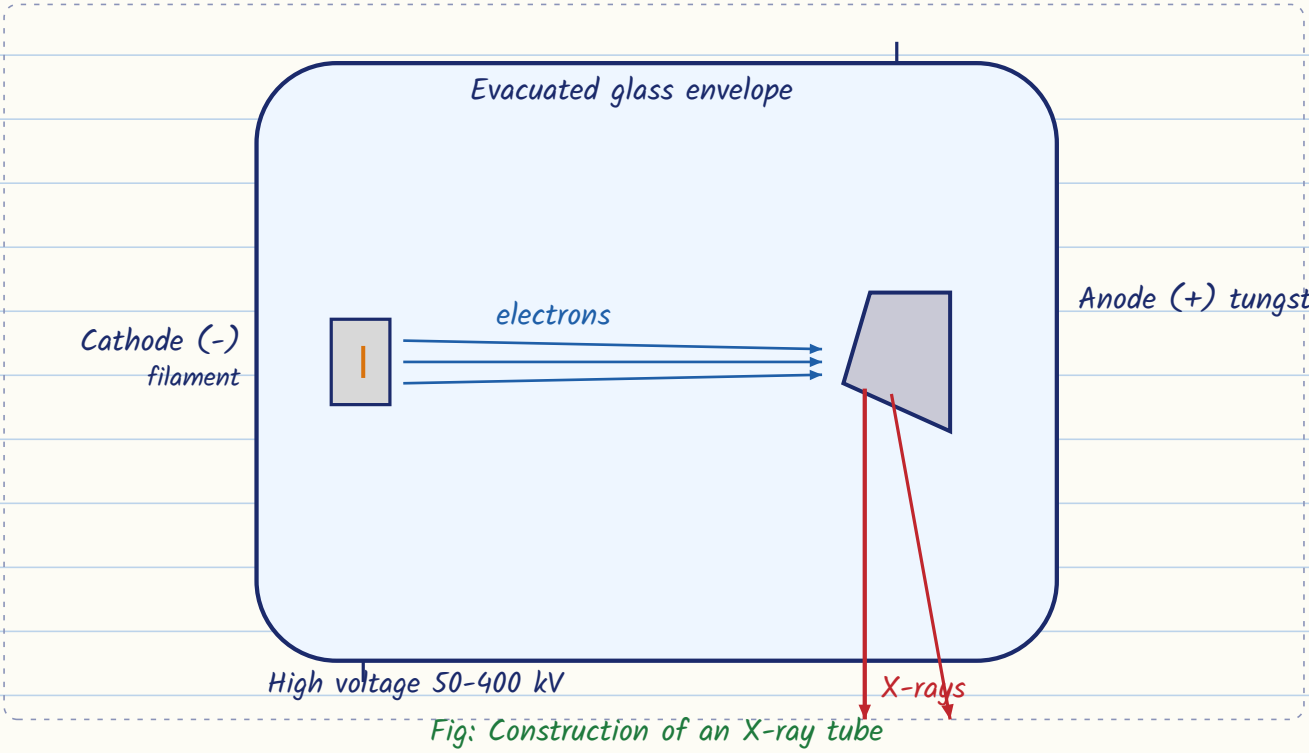


Fig: Construction of an X-ray tube

* **Cathode:** tungsten filament heated by low voltage current; emits electrons by thermionic emission. A focusing cup directs the electrons.
* **Anode:** tungsten (high melting point) target embedded in copper block (good heat conductor).

* **High voltage (kV)** between cathode and anode accelerates the electrons - kV controls **penetration** (quality).
* **Tube current (mA)** = filament heating - controls **quantity / intensity** of X-rays.
* **Vacuum envelope** of glass / metal prevents oxidation of filament and collisions of electrons with gas.
* **Window** of beryllium / glass allows X-rays to leave the tube.
* **Focal spot:** area of target bombarded by electrons - small focal spot gives sharper image.

4.3 Types of X-ray Spectrum

* **Continuous (Bremsstrahlung) spectrum:** electrons slowed in the field of nucleus lose energy as radiation of all wavelengths above a minimum value.
* **Characteristic spectrum:** an electron knocks out an inner shell electron of the target atom; the outer electron fills the gap and emits radiation of definite wavelength.
* Minimum wavelength: **lambda min = 12.4 / V (kV)** in Angstrom.

Key point:

Increase kV -> shorter wavelength, more penetration, lower contrast. Increase mA -> more intensity, shorter exposure time.

4.4 Other X-ray Generators

* **Portable X-ray units** for field work (up to 300 kV), tank-type oil-cooled.
* **Linear accelerator (LINAC):** accelerates electrons up to 4 - 25 MeV for very thick sections.
* **Betatron:** electrons accelerated in circular orbit by changing magnetic field; 10 - 30 MeV, used for steel up to 500 mm.
* **Van de Graaff generator:** electrostatic accelerator, 1 - 4 MeV.
* **Micro-focus X-ray units:** focal spot of a few microns for small electronic parts.

5. Sources of Gamma Rays

* Gamma rays are emitted from the **nucleus of unstable (radioactive) isotopes** during decay.
* Emission energy is fixed and characteristic of the isotope (monochromatic).
* Natural isotopes (radium, radon) were used earlier - now **artificial isotopes** made in nuclear reactors by neutron bombardment.
* **Activity** is measured in Curie (Ci) or Becquerel (Bq): 1 Ci = 3.7 x 10^10 Bq.
* **Half-life:** time for activity to fall to half of its value. Activity: **A = A0 e^(-0.693 t/T)** .
* **Specific activity:** activity per unit mass (Ci per gram) - high value gives small source size and sharper image.

Common Industrial Isotopes

Isotope	Half-life	Energy (MeV)	Steel thickness
Iridium-192 (Ir-192)	74 days	0.31 - 0.61	10 - 75 mm
Cobalt-60 (Co-60)	5.3 years	1.17 and 1.33	50 - 150 mm
Cesium-137 (Cs-137)	30 years	0.66	20 - 90 mm
Thulium-170 (Tm-170)	128 days	0.084	up to 10 mm (thin parts)
Selenium-75 (Se-75)	120 days	0.12 - 0.40	5 - 30 mm

X-ray vs Gamma Ray Source

Point	X-ray	Gamma ray
Origin	Electron stopping in target	Nucleus decay of isotope
Energy	Variable (kV adjustable)	Fixed for the isotope
Power supply	Needed	Not required (portable)
Switch off	Possible	Cannot be switched off
Intensity	High, controllable	Low, decays with time
Cost	High equipment	Low, but source replaced
Image quality	Better contrast	Slightly lower contrast

6. Interaction of X and Gamma Rays with Matter

* When radiation passes through matter it loses energy by **absorption and scattering**, the beam is **attenuated**.
* Narrow beam: **$I = I_0 e^{(-\mu x)}$**; Half Value Layer (HVL) = 0.693 / mu – thickness that reduces intensity to half.

6.1 Photoelectric Effect

* Photon is **completely absorbed** by an inner-shell electron, which is ejected as a photoelectron.
* Dominant at **low energy** (below about 0.1 MeV) and in high atomic number materials (Z^3 dependence).
* Gives high contrast in radiograph.

6.2 Compton Scattering

* Photon strikes a loosely bound outer electron, gives up part of its energy and is **scattered in a new direction** with longer wavelength.
* Dominant in the **medium energy range** (0.1 – 5 MeV); nearly independent of Z.
* Scattered radiation spoils image quality (fog), so lead screens and masks are used.

6.3 Pair Production

* Photon of energy more than **1.02 MeV** passes near a nucleus and converts into an **electron-positron pair**.
* Becomes important above about 5 MeV (high-energy LINAC / betatron).

6.4 Rayleigh (Coherent) Scattering

* Low energy photons are scattered without energy loss; minor effect in radiography.



Fig: Dominant interaction vs photon energy

Remember:

Total attenuation coefficient mu = tau (photoelectric) + sigma (Compton) + kappa (pair production).

7. Radiographic Equipment

7.1 Radiation Sources

* X-ray machines (portable 100 - 300 kV, fixed 400 kV, MeV units) and isotope projectors (exposure devices / cameras).

7.2 Gamma Ray Projector (Exposure Device)

* Heavy shielding container of depleted uranium / lead with an **S-shaped tube** inside which the source capsule is stored.
* Source is pushed out with a **drive cable and crank** through a guide tube to the collimator position, then retracted.
* Has a lock, warning label, and safety interlock.

7.3 Film and Cassettes

* **Industrial X-ray film:** silver bromide emulsion coated on both sides of a transparent base.
* **Film speed:** fast film = coarse grains, short exposure, poor definition; slow film = fine grain, long exposure, better sharpness.
* **Intensifying screens:** lead foil screens (front 0.05 - 0.15 mm, back 0.1 - 0.25 mm) increase blackening by emitting electrons and absorb scattered radiation. Fluorescent screens (calcium tungstate) used in medical type.
* **Cassette:** light tight flexible holder pressing film and screens together.

7.4 Film Processing

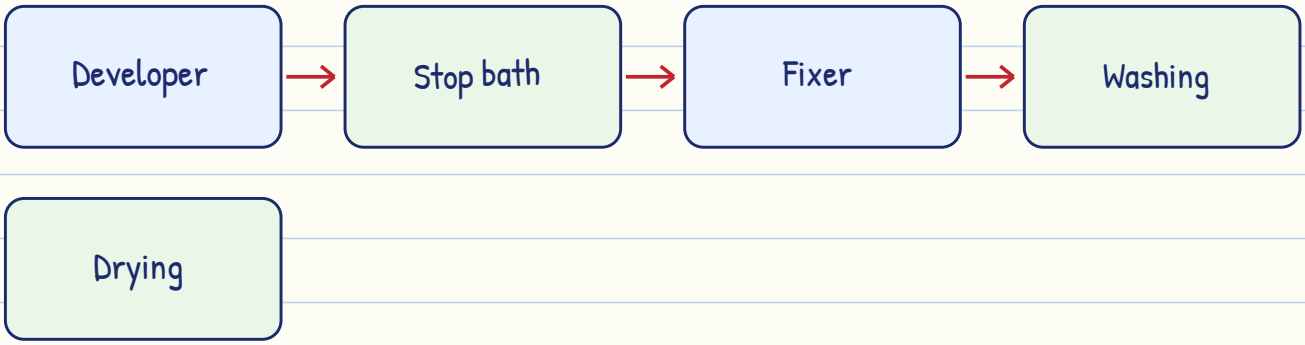


Fig: Darkroom processing sequence

* Developer converts exposed silver halide into black metallic silver; stop bath halts development; fixer removes unexposed silver bromide; washing removes chemicals.

7.5 Radiographic Image Quality

* **Radiographic density (blackening)** $D = \log10 (10 / It)$ - the usual range for acceptance is **1.8 to 4.0**.
* **Contrast:** difference in density between two areas; depends on subject contrast and film contrast.
* **Definition (sharpness):** clearness of edges; **unsharpness** $Ug = F \times t / d$ (F = focal spot size, t = object to film distance, d = source to object distance).
* **Sensitivity:** smallest flaw detectable - checked with **IQI / penetrameter** (wire type or hole type). Sensitivity = (smallest visible step / thickness) x 100 (%).

7.6 Other Detection Systems

* **Fluoroscopy / Real-time radiography:** fluorescent screen with image intensifier - instant view, used for moving parts.
* **Digital radiography (DR, CR):** imaging plates or flat panel detectors, image on computer, no chemicals.
* **Computed tomography (CT):** cross-section slices of part by many projections.
* **Xeroradiography:** selenium plate with electrostatic image.

8. Radiographic Techniques

8.1 Single Wall Single Image (SWSI)

* Source on one side, film on the other, only one wall of object is viewed. Best technique for plates and large pipes; source outside / inside.

8.2 Double Wall Single Image (DWSI)

* Both walls are penetrated but only the **wall near the film is sharply imaged** (source close to the pipe). Used for small bore pipes more than 90 mm.

8.3 Double Wall Double Image (DWDI)

* Both walls appear on the film as an **ellipse** (source offset) - for pipes less than 90 mm outer diameter. Two exposures at 90 degrees are taken.

8.4 Panoramic Technique

* Source at the centre of a cylinder / sphere; film wrapped all around – *full circumferential weld in one exposure* and saves time.

8.5 Other Techniques

* *Stereoradiography:* two shifts of source give 3-D depth location of flaws.
* *Flash / Real-time radiography* for production lines.
* *Source to film distance (SFD):* kept so that geometric unsharpness is limited; increase SFD for better sharpness (inverse square law).
* *Inverse square law:* I1 / I2 = (d2 / d1)^2 – intensity falls with square of distance.
* *Exposure time:* decided by exposure chart (thickness, kV, mA, distance) or isotope exposure calculator (activity, thickness).

Radiography rule:

Exposure = Intensity (mA or Ci) x Time. Reduce kV for contrast, increase distance for sharpness, use lead screens to cut scatter.

8.6 Interpretation of Radiographs

Flaw	Appearance on film
Porosity	Round dark spots (single or cluster)
Slag inclusion	Irregular dark lines or blobs
Crack	Fine, dark, jagged line – most serious
Lack of fusion	Straight dark line along weld edge
Incomplete penetration	Dark line at root centre
Undercut	Dark line along the toe
Tungsten inclusion	Bright (white) spot – denser than steel

* Viewing is done in a *darkened room with a high intensity film viewer (illuminator)* ; acceptance as per code (ASME Sec. V, IS 2595, etc.).

Advantages & Limitations of Radiography

* *Advantages:* detects internal flaws; gives *permanent record* ; suitable for metals and non-metals; shows nature of flaw.

* **Limitations:** planar flaws (cracks) at wrong angle can be missed; **radiation hazard**; costly; needs access to both sides; thick sections need high energy; slow.

9. Safety Aspects of Industrial Radiography

* Ionising radiation causes **biological damage** to cells - may cause burns, cataract, cancer, genetic effects. Effects are cumulative.

9.1 Units of Radiation

Quantity	Unit
Activity	Becquerel (Bq), Curie (Ci)
Exposure	Roentgen (R), C/kg
Absorbed dose	Gray (Gy) = 1 J/kg; rad (100 rad = 1 Gy)
Equivalent / effective dose	Sievert (Sv); rem (100 rem = 1 Sv)

9.2 Dose Limits (ICRP / AERB)

* Radiation worker: **20 mSv per year averaged over 5 years (max 30 mSv in any one year)**.
* Members of the public: **1 mSv per year**.
* Pregnant worker: abdomen dose limit 1 mSv for the rest of pregnancy.

9.3 Three Basic Principles - TDS



Fig: Time - Distance - Shielding

* **Time:** dose is proportional to time of exposure.
* **Distance:** dose falls with the square of distance (inverse square law).
* **Shielding:** lead, steel, concrete, depleted uranium absorb radiation. Concrete about 15 cm per HVL-based design; lead for portable shielding.

9.4 Safety Measures

* Establish **controlled and supervised area** with barriers, rope and **radiation warning signs (trefoil)**, red light / alarm during exposure.
* Personnel monitoring: **film badge, TLD (thermoluminescent dosimeter), pocket dosimeter**; direct reading alarm ratemeter.

* Area survey with **Geiger-Muller counter / survey meter** before and after exposure.
* Only **trained and licensed** radiographers (AERB certification in India); regular medical check-ups.
* Proper storage of sources in locked bunkers; **source leak test** and record keeping; transport as per rules.
* Emergency procedure for **stuck source** - evacuate, cordon, inform Radiation Safety Officer (RSO).
* Use of **collimators and lead masks** to limit beam; use of enclosed **exposure rooms with interlocks** .

Summary:

RT = best for internal flaws and permanent record; X-ray for controllable, gamma for portable; interaction by photoelectric, Compton, pair production; always follow TDS and monitoring for safety.

Important Short Questions

* Define NDT. List its advantages.
* Differentiate X-rays and gamma rays.
* Explain Compton scattering and pair production.
* What is half-life? Name any four isotopes used in RT.
* State the three principles of radiation protection.

UNIT - II

Ultrasonic Testing (UT)

1. Introduction

* Ultrasonic testing uses high frequency sound waves (0.5 - 25 MHz) to detect internal and surface flaws, measure thickness and find material properties.
* Sound above the human hearing limit (above 20 kHz) is called ultrasound .
* A short pulse of sound is sent into the part; the reflected (echo) signal tells the position and size of the flaw.
* Needs access to only one side of the part (pulse-echo) - big advantage over radiography.

Frequency Ranges of Sound

Type	Frequency	Remark
Infrasonic	below 20 Hz	not audible
Audible	20 Hz - 20 kHz	human ear
Ultrasonic	above 20 kHz	used in NDT (0.5 - 25 MHz)

Basic Principle

* Sound waves travel through a homogeneous medium at constant velocity and get reflected at an interface (crack, void, back wall) where the acoustic property changes.
* Time of flight of echo gives the distance: $d = v\,t / 2$ (v = velocity of sound, t = total round trip time).
* Amplitude of echo gives an idea of flaw size .

Pulser sends electrical pulse → Probe converts to ultrasound → Sound travels in test piece → Echo from flaw / back wall

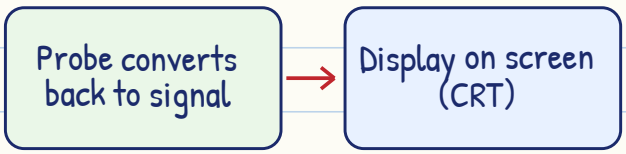


Fig: Working of pulse-echo ultrasonic testing

2. Principle of Wave Propagation

* A wave is a **disturbance** that transfers energy through a medium without permanent movement of particles.
* **Wavelength (lambda):** distance between two successive crests.
* **Frequency (f):** number of cycles per second, in Hz.
* **Velocity (v):** $v = f \times lambda$; depends on the material, not on frequency.
* **Time period** $T = 1/f$; **Amplitude:** maximum particle displacement.

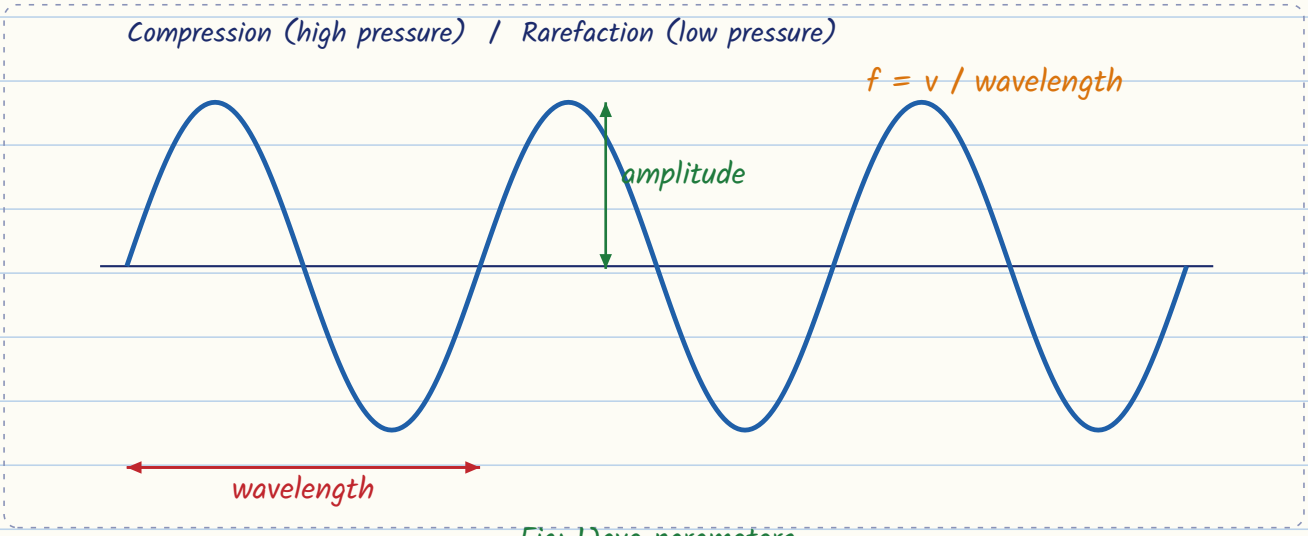


Fig: Wave parameters

Modes (Types) of Ultrasonic Waves

* **Longitudinal (compression) wave:** particle motion is **parallel** to direction of wave travel. Travels in solids, liquids and gases. Fastest wave. Used in straight-beam testing.
* **Transverse (shear) wave:** particle motion is **perpendicular** to wave direction. Only in solids. Velocity is about **half** of longitudinal velocity.
* **Surface (Rayleigh) wave:** elliptical particle motion, travels along the surface to a depth of about **one wavelength**. Detects surface and near-surface cracks.
* **Plate (Lamb) wave:** in thin plates (thickness less than a few wavelengths); symmetric and asymmetric modes.

Wave	Particle motion	Medium	Velocity
Longitudinal	Parallel to propagation	Solid, liquid, gas	Highest (steel: 5900 m/s)
Shear	Perpendicular	Solid only	About 0.5 x longitudinal (3200 m/s)
Surface (Rayleigh)	Elliptical	Surface of solids	About 0.9 x shear (2950 m/s)
Lamb	Complex	Thin plates	Depends on thickness, frequency

Velocity of Sound in Materials

Material	Longitudinal velocity (m/s)
Steel	5900
Aluminium	6300
Copper	4700
Water	1480
Air	330
Perspex	2700

* **Longitudinal velocity formula:** v = sqrt(E (1 - nu) / [rho (1 + nu)(1 - 2 nu)]) where E = Young's modulus, rho = density, nu = Poisson's ratio.
* **Shear velocity:** v = sqrt(G / rho) where G = shear modulus.

3. Acoustic Impedance, Reflection and Transmission

* **Acoustic impedance** Z = rho x v (density x velocity). Unit: kg / m^2 s (Rayl).
* When a wave meets an interface of two media with different impedance, part is **reflected** and part is **transmitted**.
* **Reflection coefficient:** R = (Z2 - Z1)^2 / (Z2 + Z1)^2 ; Transmission coefficient T = 1 - R = 4 Z1 Z2 / (Z1 + Z2)^2.
* Greater the difference in Z -> larger the reflection.
* **Steel / air interface:** almost 100% reflection (R about 0.99998) - so air gap between probe and part must be removed with a **couplant**.

* **Steel / water:** about 94% transmitted - water is a good coupling medium.
* A crack or void filled with air gives a **very strong echo** .

Remember:

Why couplant? Air gap between probe and rough surface reflects almost the entire energy. Couplants: water, oil, grease, glycerine, gel, cellulose paste.

4. Reflection, Refraction, Diffraction

4.1 Reflection

* At normal incidence the wave goes back along the same path. **Angle of incidence = angle of reflection** for oblique incidence.
* Used in pulse-echo method. Smooth flat interface -> specular reflection; rough surface scatters the sound.

4.2 Refraction - Snell's Law

* Sound passing obliquely from one medium to another **bends** due to the change in velocity.
* **Snell's law:** sin(theta1) / sin(theta2) = v1 / v2.
* Used to produce **angle beam (shear wave) probes** - Perspex wedge on steel at 45, 60, 70 degrees.

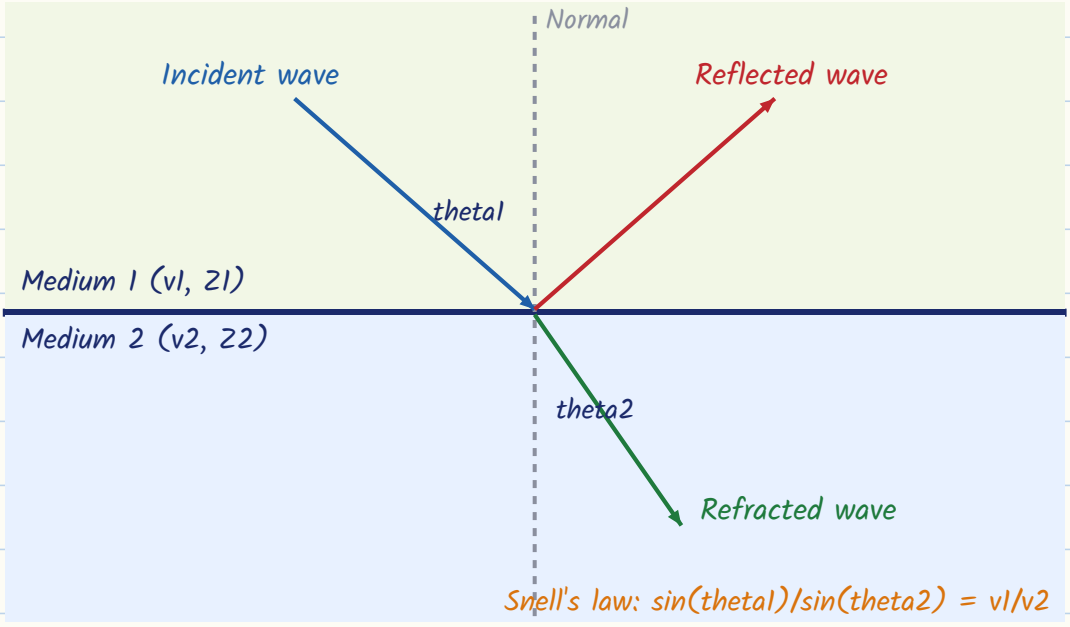


Fig: Reflection and refraction at an interface

4.3 Critical Angles

* **First critical angle:** incident angle at which the refracted longitudinal wave becomes 90 degrees (grazing) - beyond it, no longitudinal wave in the second medium.
* **Second critical angle:** incident angle at which the refracted shear wave becomes 90 degrees - beyond it, a surface wave is generated.
* Between the first and second critical angle, only a **pure shear wave** is in the test piece - the principle of the angle probe.
* For Perspex to steel: first critical angle about 27.5 deg and second about 57 deg.

4.4 Diffraction

* Bending of waves around the edge of an obstacle / small discontinuity.
* If flaw size is **much smaller than the wavelength**, the wave passes around it - not detected. So the **smallest detectable flaw is about lambda / 2**.
* Used in **TOFD (Time of Flight Diffraction)** technique - the tips of the crack emit diffracted signals which give the crack height accurately.

5. Mode Conversion

* When a wave strikes an interface **obliquely**, part of the energy changes into other wave type; for example a longitudinal wave can produce both reflected and refracted **shear** waves.
* Number and angle of waves follow Snell's law for each mode.
* Mode conversion is **useful** for making shear probes, but **harmful** when it gives false (ghost) echoes.
* In a straight beam test, longitudinal wave hitting the wall of the specimen at an angle can create shear waves and cause **spurious indications**.
* Normal incidence -> no mode conversion.

6. Attenuation

* **Attenuation** = decrease of sound intensity with distance travelled.
* Causes:
  - **Absorption** - conversion of sound energy into heat (viscous friction).

- Scattering - from grain boundaries, inclusions, porosity (strong in cast iron, austenitic steel, castings).
- Beam spreading (divergence) - geometric loss.

* Attenuation coefficient in decibel: alpha = (1/x) x 20 log10 (A1 / A2) dB per unit length.
* Intensity: I = I0 e^(-alpha x) .
* Attenuation rises with frequency and with coarse grain size - so low frequency (0.5 - 2 MHz) is used for coarse grained castings and high frequency (5 - 10 MHz) for fine grained forgings.
* Decibel (dB): gain control in the instrument; dB = 20 log10 (A2 / A1) . A change of 6 dB = double amplitude, 20 dB = ten times amplitude.

7. Sound Field of the Transducer

* Sound beam from a disc shaped crystal of diameter D has two zones:

Near Field (Fresnel zone)

* Region close to the probe with large pressure variations (maxima and minima due to interference) - flaw evaluation is unreliable here.
* Length of near field: N = D^2 / (4 lambda) = D^2 f / (4 v) .

Far Field (Fraunhofer zone)

* Beyond N, the beam diverges and intensity decreases smoothly with distance (inverse square law); testing is reliable here.
* Beam divergence (half angle theta): sin(theta) = 1.22 lambda / D - smaller crystal or lower frequency gives wider spread.
* Higher frequency or bigger crystal -> longer near field and narrower beam .

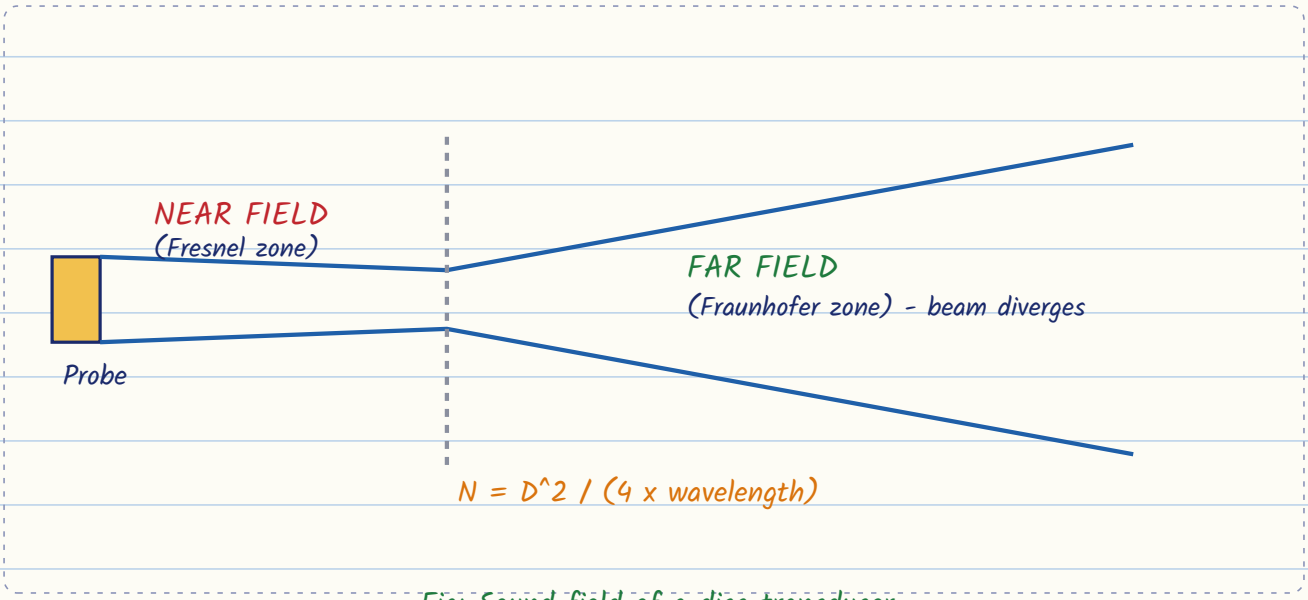


Fig: Sound field of a disc transducer

Remember:

Large diameter + high frequency = long near field, narrow beam (good directivity). Small diameter + low frequency = short near field, wide beam.

8. Piezo-electric Effect

* Certain crystals produce **electric charge when mechanically stressed** (direct effect) and **change in dimension when an electric field is applied** (inverse / converse effect).
* **Transmitter:** high voltage electrical pulse -> crystal vibrates -> ultrasound (inverse effect).
* **Receiver:** returning echo -> crystal compressed -> voltage (direct effect).
* Thickness of the crystal for resonance: $t = \lambda / 2 = v / (2f)$ - thin crystal gives high frequency.

Piezoelectric Materials

Material	Features
Quartz (natural)	Stable, hard, low efficiency; used for high frequency
Lithium sulphate	Good receiver, easily damped, water soluble
Barium titanate	High efficiency transmitter (ceramic)
Lead zirconate titanate (PZT)	Very efficient, most common, polarised ceramic
Lead metaniobate	Heavily damped, broad band, good resolution

PVDF (polymer film)	Wide bandwidth, flexible, used for special probes

* Curie temperature: above it the crystal loses piezo-electric property.

9. Ultrasonic Transducers (Probes) and Characteristics

9.1 Construction of a Probe

* **Piezoelectric element (crystal)** - the active part.
* **Backing (damping) block** - tungsten-epoxy; absorbs backward waves and **shortens the pulse** for good resolution.
* **Wear plate / protective face** - protects the crystal; matching layer improves transmission.
* **Housing and connector (BNC / Lemo)** with electrical matching network (tuning coil).

9.2 Types of Probes

* **Normal (straight beam) probe:** longitudinal wave, perpendicular to the surface. Detects laminations, forgings, thickness.
* **Angle beam probe:** crystal on a wedge; 35, 45, 60, 70 degrees; shear wave. Used for welds.
* **Dual crystal (TR) probe:** separate transmitter and receiver crystals, each tilted slightly - good for **thin sections and near surface flaws, corrosion thickness gauging** .
* **Immersion probe:** used under water with automatic scanning; may have focusing lens.
* **Focused probe:** lens gives narrow beam at a focal point - high sensitivity for small flaws.
* **Delay line probe:** plastic delay between crystal and part - for thin materials.
* **Phased array probe:** many small elements fired by software with delays to steer and focus the beam.
* **EMAT:** electro-magnetic acoustic transducer - needs no couplant; works on hot or rough surface.

9.3 Probe Characteristics

* **Sensitivity:** ability to detect small flaws (smallest echo above noise).
* **Resolution:** ability to separate two close echoes - short pulse, high frequency, good damping. Near surface resolution = **dead zone** reduction.
* **Frequency:** nominal frequency in MHz; bandwidth = (f upper - f lower) / f centre.
* **Beam spread, near field length and focal length.**
* **Damping / Q factor:** highly damped = broad band, good resolution, lower sensitivity.
* **Dead zone:** area below the surface where flaw cannot be detected due to the ringing of transmitter pulse.

Frequency	Result
Low (0.5 - 2 MHz)	More penetration, less attenuation, poor resolution - castings
Medium (2 - 5 MHz)	General purpose - welds, forgings
High (5 - 25 MHz)	Less penetration, better sensitivity - thin, fine grained

10. Ultrasonic Equipment

10.1 Flaw Detector Block Diagram

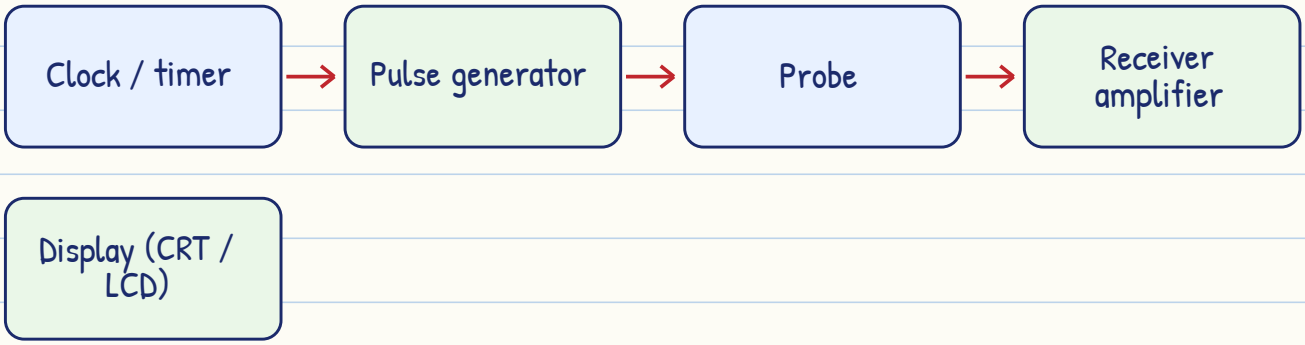


Fig: Basic block diagram of a flaw detector

* **Pulse generator:** produces short high voltage pulses (100 - 500 V) at PRF = pulse repetition frequency (50 - 2000 pulses per second).
* **Time base (sweep) generator:** moves the beam across the screen - horizontal axis = distance.
* **Receiver / amplifier:** amplifies weak echo, with gain control in dB, reject (suppression), and DAC (distance amplitude correction).
* **Display:** vertical axis = echo amplitude.

* **Gate and alarm:** monitoring section for automatic detection.

10.2 Display Types

* **A-scan:** amplitude vs time (distance) - most common; gives flaw depth and relative size.
* **B-scan:** cross-sectional view (depth vs probe position) - side view of the flaw.
* **C-scan:** plan view of the part (like X-ray) - amplitude / depth colour map by scanning x-y; used in composites.
* **D-scan, S-scan (sector scan in phased array)** .

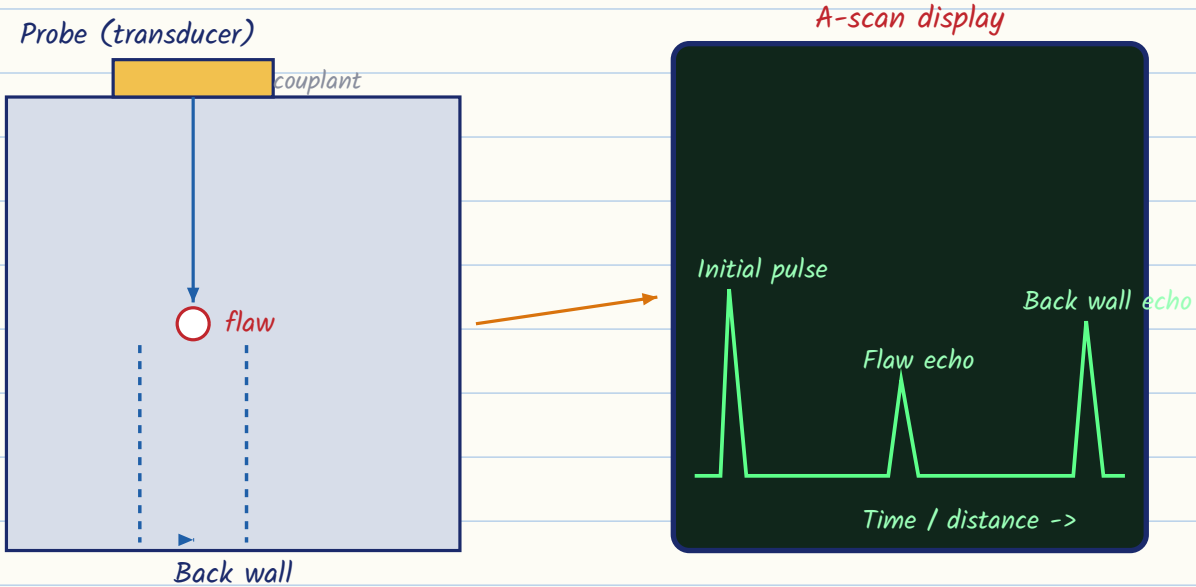


Fig: Pulse-echo method and A-scan display

10.3 Testing Techniques (Methods)

* **Pulse-echo (reflection) method:** one probe sends and receives. Most used.
* **Through transmission:** two probes on opposite sides; a flaw reduces the received signal (shadow). Needs access to both sides; no dead zone; used in automated testing.
* **Pitch-catch (two probe) technique:** separate sending and receiving probes on the same side.
* **Resonance method:** continuous wave frequency varied to find resonance -> thickness measurement.
* **Contact testing:** probe in direct contact with a thin couplant film.

* **Immersion testing:** part in a water tank; automatic scanning, better coupling, C-scan; interface echo present.
* **Gap (squirter) testing** using a water jet.

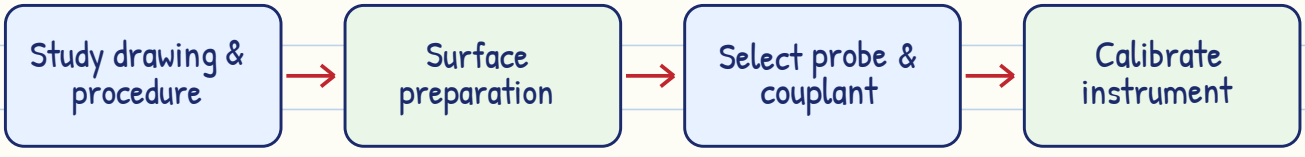
10.4 Calibration Blocks

* **V1 (IIW) block** – calibration of time base, probe index and beam angle, sensitivity.
* **V2 block** – miniature field block for angle probes.
* **Step wedge blocks** – thickness; **DAC reference blocks** with side drilled holes; ASTM basic block, area-amplitude and distance-amplitude blocks.
* Calibration is done before every test and re-checked periodically.

11. Variables Affecting Ultrasonic Test

* **Frequency and wavelength** – control sensitivity, penetration, resolution.
* **Probe diameter and type, probe angle** – control beam spread and near field.
* **Material properties:** velocity, grain size, attenuation, density, anisotropy (weld metal).
* **Surface condition:** rough surfaces, scale, paint reduce coupling (loss of echo). Use grinding, couplant of high viscosity.
* **Couplant** thickness and type.
* **Flaw characteristics:** size, shape, orientation (the best echo when flaw is **perpendicular** to beam), depth, roughness, acoustic impedance of the flaw.
* **Test piece geometry:** curvature, thin sections, edges (mode conversion, side-wall echoes).
* **Instrument settings:** gain, reject, PRF, range, pulse energy, damping, filters.
* **Temperature** – velocity changes with temperature; hot parts need special probes.
* **Operator skill** and procedure.

12. Ultrasonic Testing Procedure



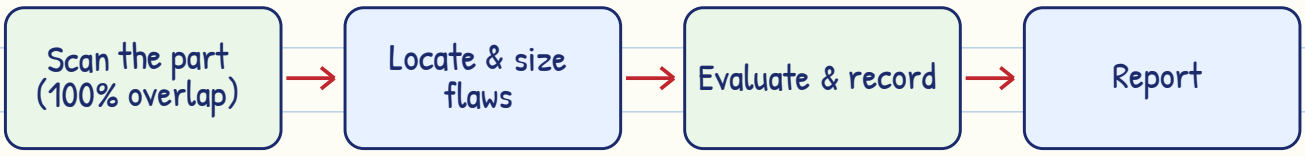


Fig: Steps in UT

* **Weld testing:** angle probes of 45, 60, 70 degrees scan the weld from both sides with zig-zag movement, using skip distance and half skip distance. Position of flaw: depth = s cos(theta), surface distance = s sin(theta), where s = sound path length.
* **Plate / forging testing:** normal probe, grid scanning to find laminations and inclusions.
* **Thickness measurement:** dual probe or digital thickness gauge (corrosion monitoring in pipes, tanks).
* **Flaw sizing techniques:** 6 dB drop (half amplitude) method for large flaws, 20 dB drop method, DGS / AVG diagram, DAC curve, TOFD.

13. Interpretation and Guidelines for Acceptance / Rejection

* Indication evaluated by **amplitude** (compared with a reference level such as DAC or a reference reflector), **length** (probe movement between the drop points), **depth and location** , and **type (planar or volumetric)** from the echo behaviour.
* Dynamic echo pattern: a **smooth, sharp, steady echo** that rises and falls rapidly with probe movement = planar flaw such as crack; a **wider echo that moves in many peaks** = porosity or slag.
* A crack gives a strong echo from one direction only; porosity gives a small echo from many directions.
* Acceptance as per codes: ASME Sec. V and VIII, ASME B31.3, AWS D1.1, API 1104, IS 4260 / IS 7666, ISO 11666.
* **Typical acceptance criteria:** all indications showing **cracks, lack of fusion or incomplete penetration are rejected regardless of size** ; others rejected when amplitude exceeds the reference level (DAC) and length exceeds the permitted value (for example one-third of weld thickness for a thin weld, up to a limit).
* Records: report with probe details, calibration block, gain, sketches, flaw position, results and name of qualified level II / III person.

Indication	Probable cause
Strong, sharp echo, moves with probe	Crack, lack of fusion
Cluster of small echoes, low amplitude	Porosity
Broad irregular echo	Slag inclusion
Echo at plate mid thickness	Lamination
Loss of back wall echo	Large defect or bad coupling / grain

14. Effectiveness and Limitations of Ultrasonic Testing

Advantages

* Very **sensitive** to planar defects such as cracks; high penetration power (several metres of steel).
* **Accurate** location and size of flaw (depth, length).
* Access to **only one surface** needed.
* **No radiation hazard** , portable equipment, instant results.
* Suitable for metals, plastics, composites, ceramics; can measure thickness and material properties (grain size, modulus).
* Automation and permanent records possible (C-scan, phased array).

Limitations

* **Skill and training** of operator is very important; interpretation is difficult.
* Surface must be **smooth and clean** ; couplant needed.
* Difficult on **rough, irregular, thin or small parts** , coarse-grained materials (cast iron, austenitic weld).
* Flaws parallel to the beam are not detected; **dead zone** near the surface.
* Needs **reference standards (calibration blocks)** .
* No permanent record in conventional A-scan.

Summary:

UT = high frequency sound, pulse-echo, uses piezo-electric probes; velocity v = f lambda; best for planar flaws; depends on couplant, impedance mismatch, attenuation and operator skill.

Important Questions

* Explain piezoelectric effect and construction of an ultrasonic probe.
* Define near field and far field. Derive N = D^2 / 4 lambda.
* Explain reflection, refraction, mode conversion and attenuation.
* Differentiate A, B and C scans. Explain pulse-echo and through-transmission methods.
* Write advantages and limitations of UT.

UNIT - III

Liquid Penetrant, Eddy Current & Magnetic Particle

PART A - LIQUID PENETRANT TEST (LPT / PT)

1. Basic Concepts

* Liquid penetrant testing detects surface-breaking discontinuities (cracks, laps, seams, porosity, cold shuts) in non-porous materials: metals, plastics, ceramics, glass.
* A coloured or fluorescent liquid with very low surface tension enters the flaw by capillary action ; after removing the excess, a developer draws the trapped liquid back out (blotting) and makes a visible indication .
* Cannot find sub-surface flaws, and cannot be used on porous materials (powder metals, wood).

Physical Principles

* Surface tension: property of liquid surface to act like a stretched skin; penetrant has low surface tension so that it spreads and wets the surface.
* Wetting ability and contact angle: good penetrant wets the surface with contact angle less than 90 degrees (nearly 0).
* Capillary action: liquid rise in a narrow gap, $h = 2\,T\cos(\theta) / (\rho\, g\, r)$ - narrower crack = higher force.
* Viscosity: low viscosity -> fast entry, but too low drains out of flaw.
* Specific gravity, volatility, flash point, chemical inertness and toxicity are other properties checked.
* Sensitivity of penetrant depends on its ability to enter fine tight cracks and to give a bright indication.

> **Remember:**
> LPT works by capillary action in the entry stage and by reverse capillary / blotting action in the development stage.

2. Liquid Penetrant System

2.1 Types of Penetrants

* **Type I – Fluorescent penetrant:** glows (yellow-green) under **ultraviolet (black) light of 320 – 400 nm**, intensity 1000 microwatt per sq cm; very high sensitivity; needs darkened area.
* **Type II – Visible (colour contrast) penetrant:** usually **red dye** against white developer; viewed in normal white light (minimum 1000 lux); good for field work.
* **Type III – Dual sensitivity:** both visible and fluorescent.

2.2 Sensitivity Levels

* Level 1/2 – ultra low / low; Level 3 – high; Level 4 – ultra high sensitivity (for very fine cracks).

2.3 Methods of Removal of Excess Penetrant (Methods A, B, C, D)

Method	Type	Removal
A	Water washable	Washed with water spray
B	Post-emulsifiable, lipophilic	Emulsifier (oil based) then water wash
C	Solvent removable	Wipe with lint free cloth moistened with solvent
D	Post-emulsifiable, hydrophilic	Pre-rinse, water-based emulsifier, final rinse

2.4 Developers

* **Dry powder** – used mainly with fluorescent penetrant.
* **Water soluble and water suspendable** – powder in water, sprayed or dipped.
* **Non-aqueous wet developer** – powder suspended in solvent, sprayed (most common for visible penetrant).
* **Plastic film developer** – special applications.
* Developer action: acts like a **blotter**, spreads the bleed-out to increase visibility, gives white **contrast background**.

2.5 Other Materials

* **Cleaners / solvents** (pre-cleaning, post-cleaning), **emulsifiers**, **remover**; aerosol cans for field kits; UV lamp and radiometer; test panels (TAM, PSM) for checking the system performance.

3. Test Procedure - LPT

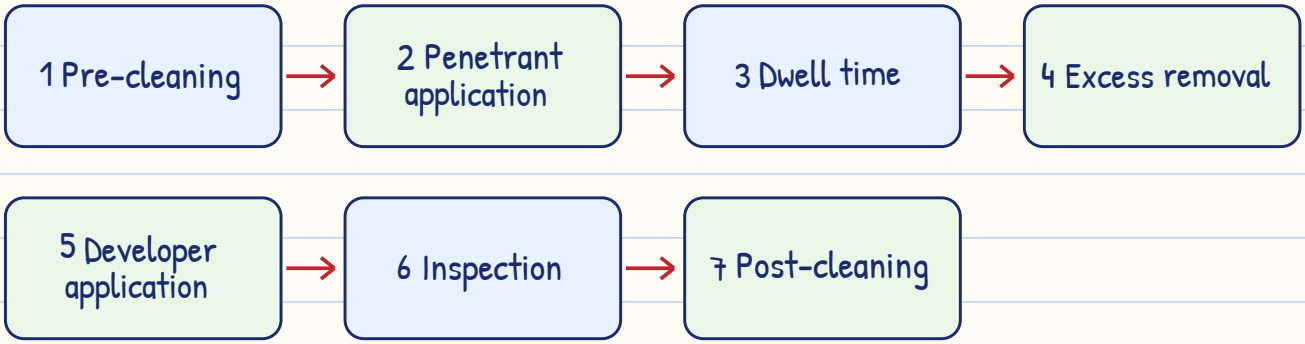


Fig: Steps of liquid penetrant testing

* **1. Surface preparation / pre-cleaning:** remove oil, grease, paint, rust, scale, dirt using solvent, vapour degreasing, detergent, ultrasonic cleaning, or wire brushing. **Do not use shot blasting** or heavy machining which smears metal over the opening of cracks. Dry the part completely.
* **2. Penetrant application:** spray, brush, dip or flood the surface (temperature 10 - 52 deg C).
* **3. Dwell time:** time the penetrant stays on the surface to enter flaws - **5 to 30 minutes** depending on material, flaw type, temperature (typical 10 - 20 min). Short for cracks in aluminium castings, longer for tight fatigue cracks.
* **4. Removal of excess penetrant:** as per method A to D; avoid over-washing (which removes penetrant from the flaw) and under-washing (leaving a high background).
* **5. Developer application:** thin uniform coat; **development time 7 - 30 minutes** (generally a minimum of 10 min).
* **6. Inspection:** under proper white light (for visible) or UV light in dark area; look for bleed-out within the development time. Flaws are **marked, recorded and measured** .
* **7. Post cleaning:** remove developer and penetrant residue to prevent corrosion.

Interpretation of Indications

Indication	Meaning
Continuous fine line	Crack, cold shut, seam
Intermittent line	Tight crack or crack partly closed
Round dots / clusters	Porosity, pin holes
Large diffuse patch	Excessive penetrant, rough surface; false indication
Wide bleed-out spread	Deep or wide flaw holds more penetrant

* **Relevant indication** from real flaws; **non-relevant indications** from part design (keyways, press fit, rough surface); **false indications** from poor processing, e.g. fingerprints, lint, penetrant on hands.
* Acceptance as per ASME Sec. V Art.6 / Sec. VIII Appendix 8, ASTM E165, E1417, IS 3658: **no linear indication longer than 1.5 mm** in most pressure vessel codes; rounded indications up to 5 mm; cracks never accepted.

4. Effectiveness and Limitations of LPT

* **Advantages:** simple, cheap, portable, high sensitivity to **fine surface cracks**, works on any non-porous shape and size, one-shot full surface coverage, easy to interpret, little training.
* **Limitations:** detects **only surface-opening flaws**; surface must be clean and smooth (not porous or very rough); not for painted / oxide covered surface; temperature range limit; time consuming steps; **no depth information**; chemical hazards and disposal of liquids; skin contact and fire hazard.

> **Tip:**
> To avoid missing tight cracks: more dwell time, ultra-high-sensitivity fluorescent penetrant, thin uniform developer and good pre-cleaning.

PART B - EDDY CURRENT TEST (ECT)

5. Principle of Eddy Current

* Based on **electro-magnetic induction (Faraday's law)**.
* An **alternating current in a coil** sets up an **alternating primary magnetic field**. When this coil is placed near a **conducting material**, circulating currents called

eddy currents are induced in it.

* Eddy currents set up a **secondary magnetic field** that opposes the primary field (Lenz's law), changing the **impedance of the test coil**.
* A **flaw, change in conductivity, permeability, dimension or heat treatment** disturbs the eddy currents -> the coil impedance (resistance and reactance) changes -> signal shown on the instrument.

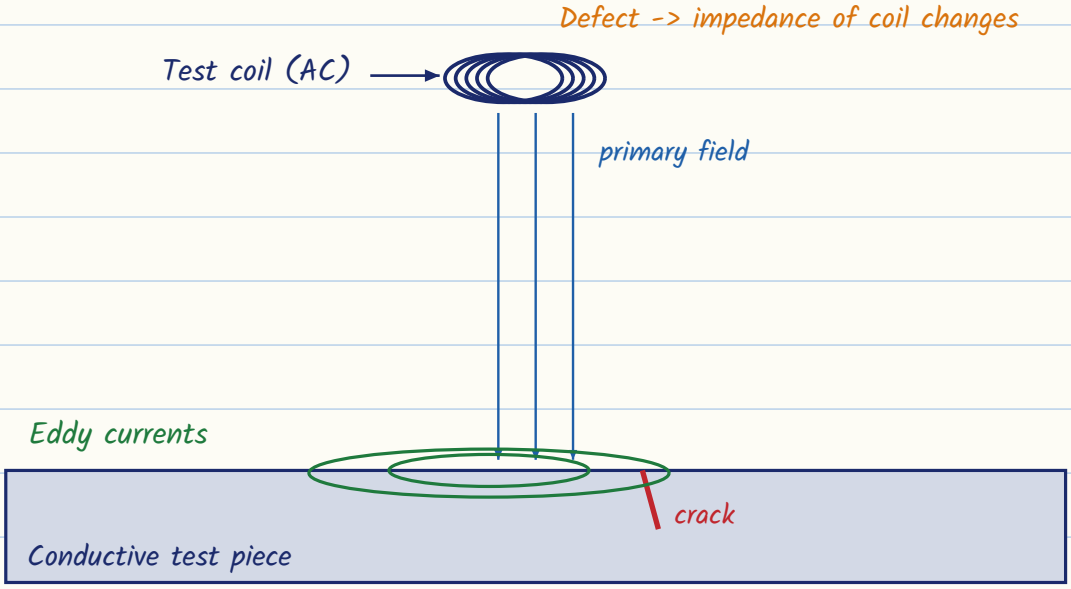


Fig: Principle of eddy current testing

Factors Affecting Eddy Currents

* **Electrical conductivity (sigma)** of the material.
* **Magnetic permeability (mu)** - ferromagnetic materials give strong noise.
* **Frequency of the test current (f)** - changes the depth of penetration.
* **Geometry** - thickness, curvature, edge effect.
* **Lift-off** - distance between the coil and the test surface.
* **Fill factor** - ratio of the specimen volume to coil volume (for encircling coils).
* **Presence of defects** - cracks interrupt the current flow.

Skin Effect and Depth of Penetration

* Eddy current density is **maximum at the surface** and falls exponentially with depth - the **skin effect**.
* **Standard depth of penetration: delta = 1 / sqrt(pi f mu sigma)** - the depth where the current density is 37% of the surface value.

* Higher frequency, conductivity or permeability -> **smaller depth of penetration** .
* Use **low frequency for sub-surface flaws** and high frequency for surface flaws; choose 3 delta as effective depth.

6. Eddy Current Test System

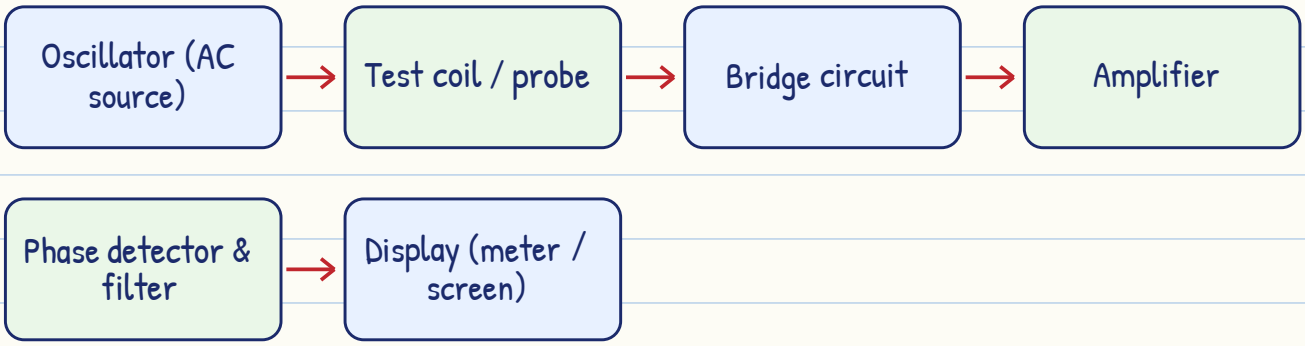


Fig: Block diagram of eddy current instrument

6.1 Probes and Coils

* **Encircling (feed-through) coil:** part passes through the coil - for tubes, wires, bars; fast testing.
* **Internal (bobbin) coil:** inserted in tubes - heat exchanger tube inspection.
* **Surface (pancake) probe:** flat coil on the surface - plates, welds, bolt holes.
* **Absolute coil:** single coil, measures total property - good for conductivity, thickness, slow changes.
* **Differential coil:** two coils compare adjacent areas - ignores gradual changes, sensitive to **small flaws** .
* **Reflection (driver-pickup) probes:** separate exciter and sensor coil.
* **Shielded probes:** ferrite shield narrows the field.

6.2 Instrument Display

* **Impedance plane diagram:** plots resistance (X-axis) against inductive reactance (Y-axis) - **phase angle indicates flaw depth** , amplitude the size; lift-off signal is separated by phase.
* Other: meter reading, alarm, strip chart, C-scan, digital data.
* Multi-frequency instruments remove unwanted signals (such as support plates).

6.3 Calibration

* Reference standard of same material with **artificial defects** – **EDM notches**, **drilled holes** of known size. Instrument gain and phase are adjusted; conductivity checked with IACS standards (International Annealed Copper Standard).

7. Applications of Eddy Current Testing

* **Flaw detection:** surface and sub-surface cracks, pits, corrosion in tubes (heat exchangers, boilers, condensers), bolt holes in aircraft structures, welds.
* **Conductivity measurement:** sorting of materials, checking **heat treatment**, alloy identification, hardness.
* **Thickness measurement:** thin metal sheets, foils, and **coating thickness** (non-conductive paint on metal).
* **Sorting** of mixed materials and of heat-treated parts.
* **Wire, rod and tube testing** in production lines at high speed.
* **Remote field ECT** for ferromagnetic tubes; **pulsed ECT** for insulated pipes; **array ECT** for fast scanning.
* **Aerospace** – fuselage lap joints, engine disks; railway wheel and rail inspection.

8. Effectiveness of Eddy Current Testing

* **Advantages:** no couplant or contact needed; **high speed** and easy automation; sensitive to **small surface cracks**; works through paint and thin coatings; portable; gives instant result; also for conductivity and thickness; works at **high temperatures**.
* **Limitations:** only for **electrically conductive materials**; shallow depth (skin effect); difficult for **ferromagnetic** materials (use saturating coil); complex signal interpretation; the **lift-off** and edge effect problems; flaws parallel to the coil winding direction may be missed; reference standards needed; geometry limits the probe selection.

Remember:

ECT = conducting material, AC coil, change in coil impedance. Depth falls as frequency increases.

PART C - MAGNETIC PARTICLE TEST (MPT / MT)

9. Magnetic Materials

* **Ferromagnetic materials** (iron, nickel, cobalt and most steels) are **strongly attracted** by magnet and can be magnetised easily; relative permeability is very high (100 to 100000).
* **Paramagnetic** (aluminium, platinum, magnesium) - weakly attracted, permeability slightly more than 1.
* **Diamagnetic** (copper, gold, silver, zinc, lead) - weakly repelled, permeability a little less than 1.
* MPT can be applied only to **ferromagnetic materials** .

Key Terms

* **Magnetic flux density B** (tesla), **field strength H** (A/m), **permeability mu = B / H** .
* **Retentivity:** ability to retain magnetism after the field is removed. **Coercive force:** reverse field needed to remove residual magnetism. **Reluctance** - opposition to flux.
* **Hysteresis loop (B-H curve):** hard magnets have a wide loop (high retentivity), soft magnets a narrow loop.

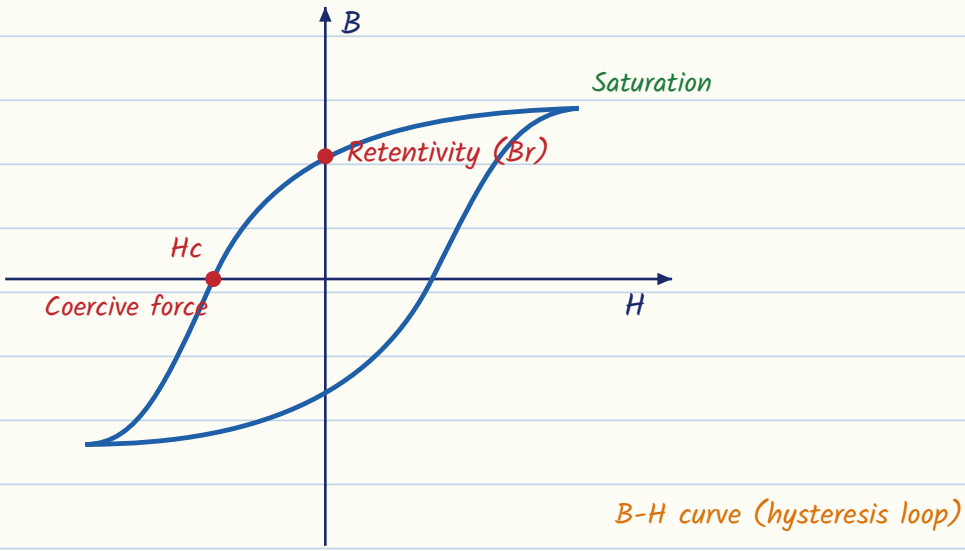


Fig: B-H curve of a ferromagnetic material

10. Magnetization of Materials

* A part is magnetised by passing **current through it** (circular field) or by **placing it in a magnetic field** (longitudinal field).
* The magnetic field must be **at 45 - 90 degrees to the flaw** for best indication. Hence two perpendicular fields are applied one after another.

Circular Magnetization

* Current passes through the part (head shot) or a central conductor (threader bar) -> circular field around the current path. Detects **longitudinal flaws** (parallel to the axis).
* **Prod method:** two hand-held electrodes (prods) pressed on the surface, current flows between them (spacing 75 - 200 mm; current 90 - 110 A per 25 mm spacing).

Longitudinal Magnetization

* Part placed in a **coil / solenoid** or between the poles of an **electromagnetic yoke** . Detects **transverse flaws** (across the axis).
* Coil: for length / diameter ratio (L/D) of 2 to 15, effective field strength depends on ampere-turns (NI).

Types of Magnetizing Current

* **AC (alternating current):** skin effect so **best for surface cracks** ; easy to demagnetise; used in portable yokes.
* **DC / rectified half wave (HWDC):** penetrates deeper, **subsurface flaws** , full wave (FWDC) smooth.
* **Pulsed / capacitor discharge** and **permanent magnets** are other sources.
* **Multi-directional magnetisation** with swinging fields for complex parts.

Methods of Magnetization

Method	Field	Flaw detected
Head shot / direct contact	Circular	Longitudinal flaws

Central conductor	Circular	Flaws on inside and outside surface of tubes
Prods	Circular	Weld flaws, large parts
Coil	Longitudinal	Transverse flaws
Yoke	Longitudinal (in between poles)	Any flaw perpendicular to the pole line, welds

II. Principle of Magnetic Particle Test

* When a ferromagnetic part is magnetised, magnetic flux lines flow through the part. A ***discontinuity (crack) perpendicular to the flux*** acts as an ***air gap of high reluctance*** , and the flux is forced out of the surface, forming a ***magnetic leakage field*** (a pair of poles).
* Fine ***ferromagnetic particles*** sprinkled on the surface are attracted to the leakage field and ***collect along the flaw*** , forming a visible indication that is wider and clearer than the flaw itself.

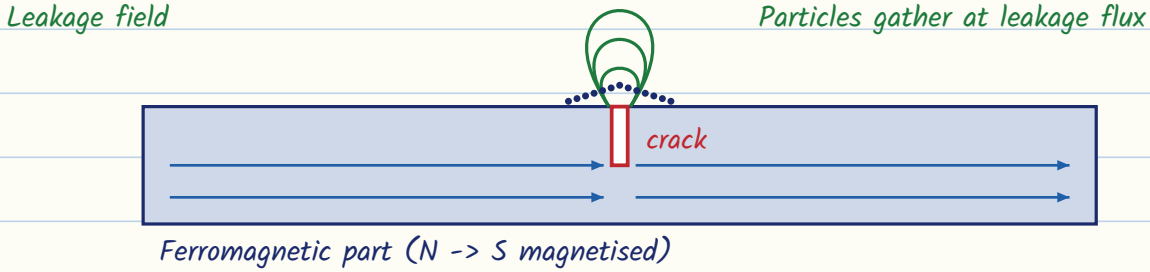


Fig: Leakage flux at a crack attracts magnetic particles

* Flaws perpendicular to the lines of flux give the best indication; ***parallel flaws are not detected*** .
* Sensitivity is highest for ***surface and near surface*** flaws (up to about 3 mm depth); decreases rapidly with depth.

Magnetic Particles

* **Dry powder:** iron oxide particles (50 – 150 micron) – coloured red, grey, black, yellow; good for rough surfaces, subsurface flaws with DC.
* **Wet particles:** fine particles (5 – 10 micron) in water or oil carrier – **higher sensitivity** for fine surface cracks, bath concentration 1.2 – 2.4 ml per 100 ml (visible), 0.1 – 0.4 ml (fluorescent).
* **Fluorescent particles** viewed under black light (UV) – best sensitivity.
* Visible particles are chosen to give **contrast** with the part (black on white lacquer background for wet).

12. Magnetic Particle Test Equipment

* **Portable electromagnetic yoke** (AC / DC) with articulated legs.
* **Prods and cable set. Permanent magnet yoke** (for places without power, explosive atmosphere).
* **Horizontal wet bench (stationary)** with head and tail stocks, coil, bath tank, pump, UV lamp and demagnetizing coil – for mass production.
* **Mobile units** , **flexible cable (coil wrapped) and threaders** .
* **Light meter and UV radiometer** , magnetic field indicator (pie gauge, Hall effect gaussmeter), and **Quantitative Quality Indicator (QQI)** shims, ASME **Ketos ring** to check system performance.

13. Magnetic Particle Test Procedure

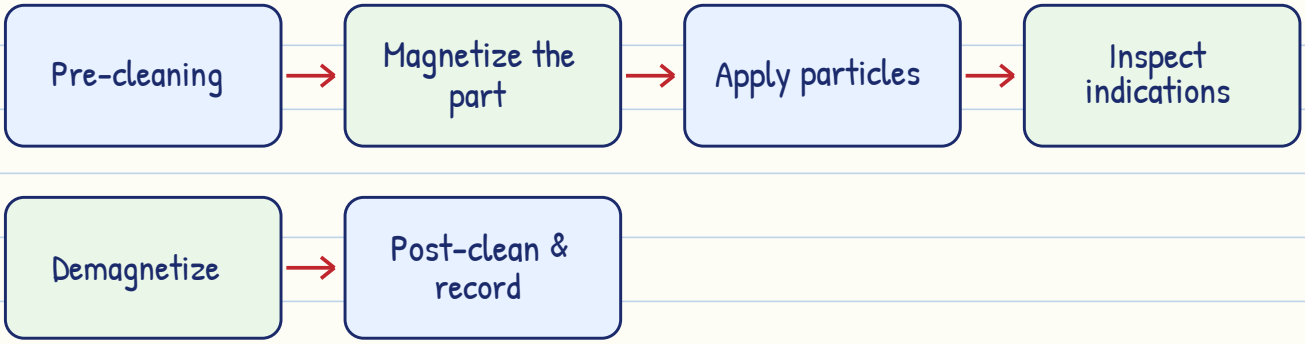


Fig: Magnetic particle testing steps

* **1. Preparation:** clean oil, scale, paint; surface within 1 mm coating is allowed with sensitivity check.
* **2. Magnetisation:** choose method and current (rule of thumb: 12 – 36 A per mm of diameter for head shot; yoke must lift 4.5 kg if AC and 18 kg if DC).

* **3. Application of particles: continuous method** (particles applied while the current flows - recommended, more sensitive) or **residual method** (applied after current is removed - for high-retentivity steels).
* **4. Inspection:** under white light (minimum 1000 lux) or UV light (1000 microwatt per sq cm in dark area at the surface); flaws form sharp, tightly held indications.
* **5. Demagnetisation** (below) and **6. post-cleaning and protection** against rust.

14. Demagnetization of Materials

* Part retains **residual magnetism** after testing. It can attract chips and swarf, damage bearings and gears, interfere with welding (arc blow), affect instruments, compass and subsequent machining.
* **Need for demagnetisation:** for moving parts, aircraft, welding next, plating, electronic instrument parts.
* **Method:** the part is subjected to an **alternating field of decreasing strength** , which reverses the domain direction in smaller and smaller steps until the residue is nearly zero.
  - **AC coil:** pass the part slowly through a coil with AC (about 5 - 7 A) and withdraw it 1 m or more away.
  - **DC reversing and decaying** for deep magnetised parts.
  - **Heating above Curie temperature** (770 deg C for iron) also removes magnetism.
* **Check** with a field indicator; residual field should be **below 3 gauss (0.3 mT)** .

15. Standardization and Calibration

* **System performance** verified with **test rings (Ketos ring)** having drilled holes at different depth, and test pieces with **known cracks** .
* **Field strength** checked using **pie gauge, QQI shims or gaussmeter** . Tangential field about **30 - 60 gauss (2.4 - 4.8 kA/m)** is the usual requirement.
* **Yoke lifting power test** before use (AC: 4.5 kg at maximum leg spacing, DC: 18 kg).
* **Particle concentration** checked by a centrifuge tube (settling test); **UV lamp intensity and white light level** with a calibrated meter; ammeter calibration

every six months.

16. Interpretation and Evaluation

Indication	Appearance
Surface crack	Sharp, well defined, tightly packed, continuous line
Subsurface flaw	Fuzzy, broad, less defined, usually with DC dry powder
Lamination, seam	Straight or curved lines
Non-relevant (magnetic writing, change of section)	Weak, broad, removed after demagnetisation

* ***Relevant indication*** - from a real discontinuity; ***non-relevant*** - due to design (sharp keyways, thread root); ***false*** - poor technique (particle trapped, rough weld toe).
* Evaluation as per ASTM E709, E1444, ASME Sec. V Art.7, IS 3703: cracks and linear indications over 1.5 mm usually rejected.

17. Effective Applications and Limitations of MPT

* ***Applications:*** weld inspection, castings and forgings, shafts, crank shaft, gears, bolts, railway axle, pressure vessel nozzles, aircraft landing gears, in-service fatigue crack checks.
* ***Advantages:*** fast, cheap, simple, high sensitivity to ***fine surface and near-surface cracks*** , shows flaw through thin coating, immediate result, portable, accept irregular shape.
* ***Limitations:*** ***only ferromagnetic materials*** ; flaws parallel to field not found; needs ***demagnetisation*** ; the surface must be reasonably clean; electric burns risk with prods; needs ***two directions*** for full coverage; poor depth information; large currents for big parts; paint and thick coating reduce sensitivity.

Comparison of Surface Methods

Property	LPT	MPT	ECT
Material	Non-porous	Ferromagnetic	Conductive
Flaws found	Surface-opening	Surface and slightly subsurface	Surface and near surface
Cost, speed	Low, slow	Low, fast	Medium, very fast
Contact needed	Yes (liquid)	Yes (particles)	No
Training	Low	Medium	High

Summary:

LPT - capillary action for non-porous; ECT - induced currents in conductors, change in impedance; MPT - leakage flux collects particles on ferromagnetic parts; MPT needs demagnetisation.

Important Questions

* Explain the steps of liquid penetrant testing with a neat flow chart.
* Discuss the principle and applications of eddy current testing.
* Explain skin effect and depth of penetration.
* Describe magnetisation techniques and the demagnetisation process in MPT.
* Compare LPT, MPT and ECT.

UNIT - IV

Infrared & Thermal Testing

1. Introduction and Fundamentals

* Infrared & Thermal Testing (IRT) is an NDT method that uses temperature differences and thermal radiation of a surface to find flaws, hidden defects and changes in the condition of a component.
* Principle: a defect changes the flow of heat through a part; this makes a local hot or cold spot on the surface, which is detected by contact thermometers, heat sensitive coatings or an infrared camera .
* Everything above absolute zero (0 K = -273 deg C) emits infrared radiation ; hotter objects emit more.
* Used for: delamination, disbonds, cracks, voids, corrosion, moisture, blocked flow, loose electrical joints, insulation faults, thickness loss.

Basic Terms

* Temperature (T): measure of thermal state - K or deg C.
* Heat (Q): energy transfer due to temperature difference - Joule.
* Thermal conductivity (k): W / m K - ability to conduct heat.
* Specific heat (c): energy to raise 1 kg by 1 K (J / kg K).
* Thermal diffusivity (alpha) = k / (rho c) - m^2 / s - how fast heat spreads. A low-diffusivity region (a void) holds heat.
* Thermal effusivity (e) = sqrt(k rho c) - governs contact temperature of two bodies.
* Thermal resistance R = L / (k A) .

2. Infrared Radiation - Electromagnetic Spectrum

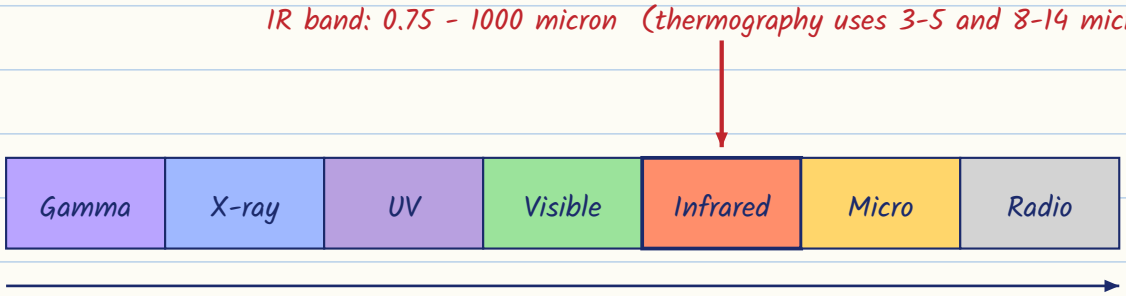


Fig: Infrared band in the electromagnetic spectrum

* Infrared region is between **visible red light (0.75 micron) and microwaves (1000 micron)** .
* Sub-bands: **Near IR (0.75 - 1.4 micron), Short wave IR (1.4 - 3), Mid wave IR (3 - 5 micron), Long wave IR (8 - 14 micron), Far IR (above 15 micron)** .
* Atmosphere is transparent only in windows **3 - 5 micron and 8 - 14 micron** , hence thermal cameras work in these bands.
* Infrared travels at light speed, in straight lines, can be reflected, refracted and absorbed.

Laws of Thermal Radiation

* **Black body:** ideal body that absorbs all radiation and emits the maximum possible at its temperature (emissivity = 1).
* **Stefan-Boltzmann law:** total energy radiated **W = epsilon sigma T^4** (sigma = 5.67 x 10^-8 W / m^2 K^4; epsilon = emissivity). Radiation increases as the **fourth power** of temperature.
* **Wien's displacement law: lambda max x T = 2898 micron K** - the peak wavelength shifts to shorter wavelength as T rises. For a body at 300 K, lambda max is about 9.7 micron (in LWIR).
* **Planck's law:** distribution of radiation over wavelength.
* **Kirchhoff's law:** good absorbers are good emitters, **absorptivity = emissivity** .

* **Emissivity (epsilon):** ratio of radiation of a real surface to that of a black body at the same temperature (0 to 1). Polished metal = 0.05 – 0.1; oxidised steel = 0.8; paint = 0.9; human skin = 0.98.
* Radiation seen by camera = **emitted + reflected + transmitted** part: **epsilon + rho + tau = 1**.

Remember:

Low emissivity shiny surface -> false low temperature reading. Cover with black paint or tape (known emissivity) or correct the emissivity setting.

3. Heat Transfer

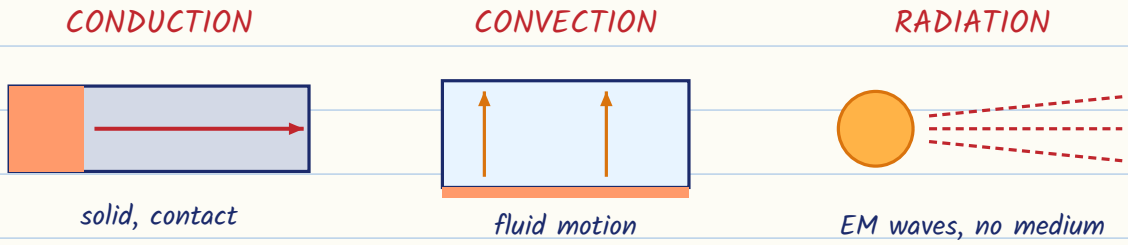


Fig: Modes of heat transfer

* **Conduction:** heat flow through a solid or between touching bodies due to temperature gradient. **Fourier's law: q = - k A dT / dx**. A flaw (air gap) has lower k and **obstructs** heat flow.
* **Convection:** heat transfer by motion of a fluid (liquid / gas). Newton's law of cooling: **q = h A (Ts - Tf)** ; natural or forced.
* **Radiation:** heat transfer by electromagnetic waves, needs no medium. **Net exchange q = epsilon sigma A (T1^4 - T2^4)**.
* In thermal NDT, **conduction inside** the part is related to flaw location and **convection and radiation at the surface** are the loss paths and the source of the signal.
* Defects:

- **Insulating** flaw (void, delamination) -> heat blocked -> **hot spot on heated side**, cooler surface behind it.
- **Conductive** flaw (metal inclusion in plastic) -> **cold spot** on heated side.
- Moisture and corrosion change heat capacity and conductivity.

4. Active and Passive Techniques

4.1 Passive Thermography

* The component is **naturally at a different temperature from its surroundings** due to operation or process. No external heating needed.
* **Examples:** hot bearings, overheated electrical joints and fuse, steam line insulation, furnace refractory, building heat loss, hot spots in PCB, moisture in roofs, human fever screening.
* Fast, simple, done on a **live equipment**, qualitative.

4.2 Active Thermography

* An **external stimulus (heating or cooling) is applied** to create a thermal contrast over defects.
* **Heat sources:** flash lamp, halogen lamp, laser, hot air, hot water, induction heating, ultrasound (vibro-thermography), microwave, Joule heating by current.
* **Modes of heating:** reflection mode (camera and source on same side) for sub-surface flaws; transmission mode (camera on opposite side) for through-thickness flaws.
* Gives **quantitative** depth and size of defects.

Point	Passive	Active
External energy	Not needed	Required (heat source)
Target	Operating equipment, in-service	Material defects, composites
Result	Qualitative, temperature anomaly	Quantitative defect depth / size
Example	Electrical panel check	Delamination of aircraft panel

5. Lock-in and Pulse Thermography

5.1 Pulse Thermography (PT)

* A short, high-energy heat pulse (flash 1 ms - few seconds) heats the surface; the IR camera records surface temperature decay with time.
* The heat diffuses inward; an insulating flaw at depth z reduces cooling rate -> appears as a warmer area after time t = z^2 / alpha (alpha = diffusivity).
* Deeper flaws appear later and with lower contrast. Gives depth from time of appearance.
* Advantages: fast (seconds), wide area, simple. Limitations: non-uniform heating, surface emissivity variation, and low depth resolution; needs high power flash.
* Processing: thermal signal reconstruction (TSR) , pulsed phase thermography (PPT) by Fourier transform.

5.2 Lock-in (Modulated) Thermography

* The surface is heated by a sinusoidally modulated source at frequency f_lock; the camera records the response at the same frequency ("locked-in").
* The amplitude and phase images are calculated; phase is independent of surface emissivity and non-uniform heating -> better detection of deep defects.
* Thermal diffusion length: mu = sqrt(alpha / (pi f)) - depth probed; lower frequency gives deeper penetration , up to about 1.8 mu depth in phase image.
* Advantages: high sensitivity, low power source, quantitative depth, deals with emissivity problem. Limitations: slower (needs several cycles), one frequency at a time.
* Variants: vibro-thermography / lock-in ultrasound.

Point	Pulse	Lock-in
Heating	Single short pulse	Periodic sine wave
Time	Seconds	Minutes
Image	Temperature vs time	Amplitude and phase

Depth range	Medium	Deeper, better phase
Equipment	Flash lamp, camera	Lamp / laser with modulator, camera

6. Contact and Non-contact Thermal Inspection Methods

6.1 Contact Methods (touch the object)

* **Thermometers:** mercury / alcohol type, bimetallic thermometers.
* **Thermocouples:** two dissimilar metals (K type chromel-alumel, J, T) produce emf (Seebeck effect); range -200 to 1300 deg C.
* **Resistance temperature detectors (RTD, platinum PT100)** - accurate, linear.
* **Thermistors** - semiconductor, sensitive but narrow range.
* **Heat sensitive paints, crayons, pellets, labels, papers, and liquid crystals** (below).
* **Limitations:** point measurement only, thermal contact error, needs access to the surface, slow response, disturbs the temperature of small objects.

6.2 Non-contact Methods

* **Radiation pyrometers / spot IR thermometers** - single point, quick, safe on moving, hot or live objects.
* **Optical pyrometer (disappearing filament)** for very hot bodies.
* **Thermal imaging (IR cameras)** - full-field 2-D temperature map.
* **Advantages:** no contact, fast, safe distance, area view. **Limitations:** depends on emissivity, reflections, atmosphere, distance.

7. Heat Sensitive Coatings and Materials

7.1 Heat Sensitive Paints

* Special paint that **changes colour at a fixed temperature** (irreversible, indicates maximum temperature reached). Each paint has a transition temperature range (about 40 to 1300 deg C). Applied by brush or spray.
* Used on **turbine blades, engine parts, electrical bus bars, furnace** , boilers.
* Types: single change and **multi-change (several colours)** paints; **thermo-chromic** reversible paints.

7.2 Heat Sensitive Papers

* Papers impregnated with chemicals (such as ones that change from white to black) at a certain temperature; used as ***temperature labels / strips*** and in thermal printers. Quick checks on parts in ***inaccessible*** locations.

7.3 Thermally Quenched Phosphors

* ***Phosphors*** glow under UV light; as the ***temperature rises, the brightness of the luminescence goes down*** ("thermal quenching") and the decay time shortens.
* A phosphor coating (such as magnesium fluorogermanate) is applied on the surface, excited with UV; the ***brightness map*** shows the temperature distribution. Used for ***non-contact surface temperature mapping*** of electronic circuit boards, turbine blades.

7.4 Liquid Crystals

* ***Organic compounds*** that exist between solid and liquid phase and ***selectively reflect light of a colour depending on temperature*** - they change colour from ***red -> yellow -> green -> blue -> violet*** with rise in temperature.
* ***Cholesteric (chiral nematic) liquid crystals*** are used; range of colour play can be fixed from -30 to 250 deg C with very narrow band 1 - 2 deg C accuracy.
* They are ***reversible*** , give full surface display and need no electronic equipment.
* ***Applications:*** detecting hot spots on printed circuit boards, bonded (adhesive) joints, voids in honeycomb, cracks in welds, delamination, medical thermography (skin temperature).

7.5 Techniques for Applying Liquid Crystals

* Surface must be cleaned and coated with ***black paint (background)*** for good colour contrast.
* ***Spraying:*** liquid crystal mixture in solvent sprayed with an airbrush on the black surface.
* ***Brush / painting*** on a flat surface.
* ***Encapsulated (micro-encapsulated) slurry or sheet:*** the crystals are in tiny capsules in a binder and applied by spraying or as ***self-adhesive sheets*** - easy to use, durable and protected from contamination.

* **Dip coating, printing** of thin films for tests of small parts.
* Viewing in white light; record with a colour camera.

7.6 Other Temperature Sensitive Coatings

* **Thermographic phosphors, temperature indicating crayons (Tempilstik),** thermochromic inks, **pyrometric cones, fusible pellets and melting point paints** (melt at exact temperature – 40 to 1400 deg C).
* **Thermal barrier coatings** check; **pressure and temperature sensitive paints** in wind tunnel tests.

8. Inspection Methods Using Infrared

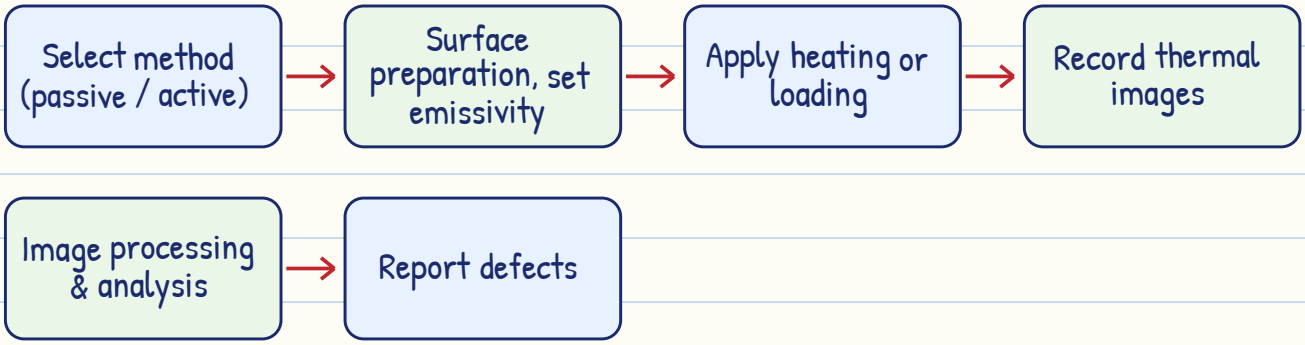


Fig: Procedure for IR thermographic inspection

* **Surface preparation:** clean surface, apply thin black matte paint if surface is shiny (to increase emissivity and avoid reflections).
* **Calibration:** camera with a **black body reference source** ; set emissivity, reflected apparent temperature, distance, humidity.
* **Select view angle and camera distance** (spatial resolution IFOV).
* **Stimulation:** thermal (flash, lamp, hot air), mechanical (ultrasound / vibration), electrical (current), or microwave / eddy current.
* **Record sequence** at suitable frame rate and duration (some seconds to minutes).
* **Image processing:** background subtraction, contrast enhancement, FFT, principal component analysis, TSR.
* **Interpretation:** compare with reference area, find the defect size, depth; report with thermogram, visual image, settings, date, ambient conditions.

9. Infrared Radiation and Infrared Detectors

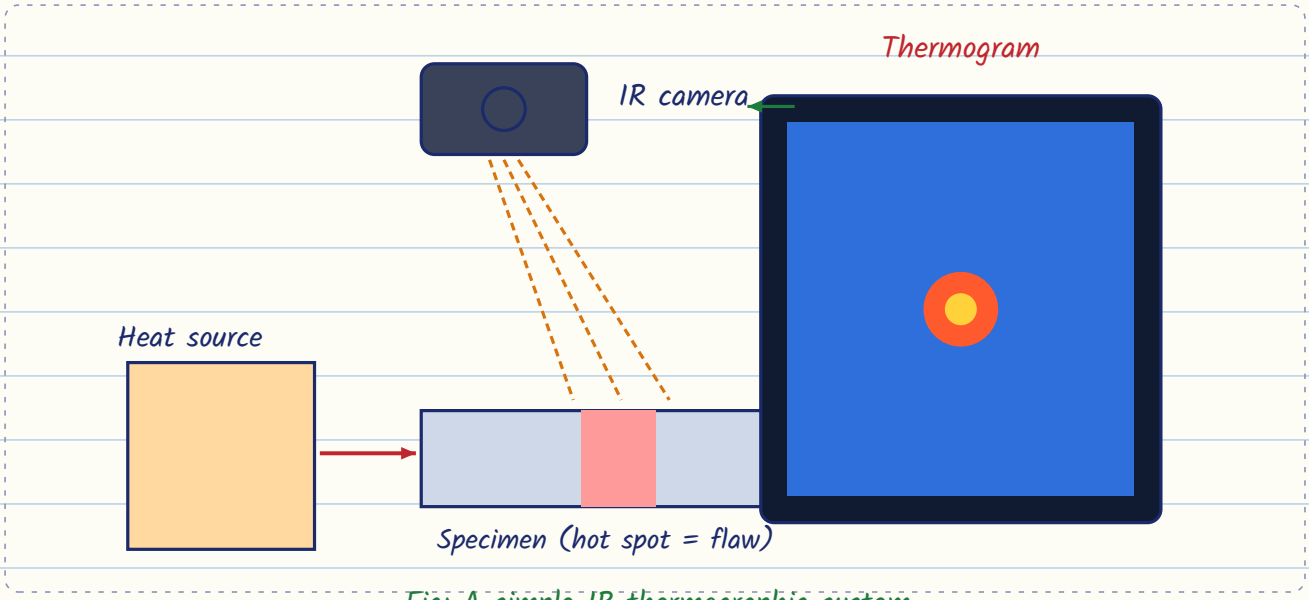


Fig: A simple IR thermographic system

9.1 Infrared Camera System

* **Optics (germanium lens)** - IR transparent materials: Ge, ZnSe, silicon, sapphire (glass does not pass LWIR).
* **Detector (focal plane array)** converts IR radiation into electrical signal.
* **Cooling system** (Stirling or liquid nitrogen for cooled detectors).
* **Signal processor, display, memory** and software.

9.2 Types of IR Detectors

* **Thermal detectors:** respond to **heating effect** of radiation; wavelength independent, uncooled, slower, cheaper.
  - **Bolometers** (microbolometer array of vanadium oxide / amorphous silicon) - resistance changes with temperature; most common.
  - **Thermocouples / thermopiles** , **pyroelectric detectors (TGS, lithium tantalate)** - charge changes with temperature.
* **Photon (quantum) detectors:** radiation frees electrons directly in semiconductor; **very sensitive and fast** , wavelength dependent; usually **cooled (77 K)** to reduce noise.
  - **InSb (indium antimonide, 3 - 5 micron), HgCdTe (MCT, 3 - 12 micron), QWIP (quantum well, 8 - 9 micron), PbS, PbSe, InGaAs** .

Point	Thermal detector	Photon detector
Response	Heat effect	Photon absorption
Speed	Slow (ms)	Fast (microseconds)
Sensitivity	Moderate (NETD 50 mK)	High (NETD below 20 mK)
Cooling	Not required	Required (cryogenic)
Cost	Low	High

* Performance terms: NETD (noise equivalent temperature difference), spatial resolution (IFOV), field of view, frame rate, spectral range, accuracy +/- 2 deg C.

10. Thermo-mechanical Behaviour of Materials

* Thermal expansion: a body expands when heated, change in length dL = alpha_L L dT (alpha_L = coefficient of linear expansion).
* When expansion is constrained, thermal stress = E alpha_L dT develops. Uneven heating -> thermal gradient -> stress, warping, cracks, fatigue (thermal cycling).
* Thermoelastic effect: when a body is elastically loaded (tension or compression), its temperature changes slightly - rise in compression and fall in tension. Thermoelastic stress analysis (TSA) uses IR camera to map the stress field: dT = - K T (sigma1 + sigma2), where K = thermoelastic constant.
* Heat generated in plastic deformation, friction and crack tip - the fatigue crack growth, bolt looseness and rubbing parts show heating detected with IR camera.
* Vibro-thermography (sonic IR): ultrasonic energy vibrates faces of a crack; friction at the crack faces generates heat so only closed cracks and delaminations appear hot.
* Creep, thermal fatigue and thermal shock are detected by monitoring temperature of boiler, turbine parts.
* Material property and heat treatment (conductivity, diffusivity) can be mapped by thermography.

11. IR Imaging Applications

11.1 Aerospace Applications

* **Composite structures (CFRP, GFRP):** detection of **delamination, disbonds, impact damage (BVID), porosity, water ingress** in wings, fuselage panels, rudders, radomes.
* **Aircraft skin:** corrosion and lap-joint disbonds, thinning, rivets; repair patch quality.
* **Engine and turbine blades:** cooling hole blockage, thermal barrier coating (TBC) delamination, cracks; space shuttle tiles / heat shield inspection.
* **Rocket and missile:** motor case bond line, propellant voids, insulation.
* Pulse thermography and lock-in used for fast checking of large panels.

11.2 Electronic Components

* **Hot spot detection** on PCB - short circuit, overloaded or failing component, poor solder joint, delamination of chips, heat sink contact.
* **Thermal design verification** of components and power devices (IGBT, LED); **failure analysis** of integrated circuits (lock-in thermography finds micro leakage).
* Switchgear, transformers, bus bars, connections and cables in **electrical substations** (predictive maintenance).

11.3 Honeycomb and Sandwich Structures

* Sandwich panels have thin **skins bonded to a honeycomb core** (Al, Nomex, foam).
* Defects: **disbond (skin-core), crushed core, water ingress, node failure, adhesive voids** .
* Pulse or lock-in thermography with transmission or reflection mode: **disbonded area retains heat and appears hot** , water-filled cells appear cooler on cooling due to high heat capacity.
* Liquid crystals also used for quick checks of honeycomb bond.

11.4 Other Applications

* Buildings (insulation and moisture), pipelines, refinery furnaces, boilers tubes, refractory lining, welding monitoring, medical thermography, forest fire, mines.

12. Case Studies

Case 1 - Delamination in a CFRP Aircraft Panel

* **Problem:** impact damage on carbon fibre wing skin, invisible on surface.
* **Method:** active pulse thermography using two flash lamps (2 ms, 6 kJ), IR camera at 100 Hz, reflection mode.
* **Result:** delamination region showed **slower cooling (warmer)** at t = 0.4 s; size and depth estimated by TSR. **Panel repaired** after the area was marked.

Case 2 - Hot Joint in an Electrical Panel

* **Problem:** periodic trips in a 415 V distribution panel.
* **Method:** passive IR survey under load.
* **Result:** a loose lug on one phase was **35 deg C above** the other phases -> torque corrected and temperature normal. Avoided fire and downtime.

Case 3 - Honeycomb Disbond in a Helicopter Blade

* **Method:** lock-in thermography at 0.1 Hz with halogen lamps.
* **Result:** phase image displayed the **skin-core disbond** as clear patches; confirmed by tap test and ultrasonic. Early detection avoided blade failure.

Case 4 - PCB Failure Analysis

* **Method:** IR camera on powered board; one capacitor showed **60 deg C hot spot** ; found a short circuit due to solder bridging.

13. Advantages and Limitations of IR Thermography

* **Advantages: non-contact** , safe, fast, full-field image, works on live equipment, remote and hazardous places, sensitive, easy to record and report, no radiation.
* **Limitations: surface temperature only** (limited depth); depends on emissivity and reflections; shiny, low emissivity metals are difficult; thick or high conductivity materials; cost of high-quality cameras; need trained thermographer; effect of wind, sun, moisture, ambient temperature; the results are affected by thermal contrast and heat sources.

Summary:

IRT detects flaws from surface temperature patterns. Passive for operating equipment, active (pulse / lock-in) for defects. Needs correct emissivity, IR camera in 3 - 5 or 8 - 14 micron window. Used widely on composites, electronics and sandwich panels.

Important Questions

* Explain the principle of thermal NDT and the laws of thermal radiation.
* Differentiate active and passive thermography; explain pulse and lock-in techniques.
* Describe heat sensitive paints, liquid crystals and thermally quenched phosphors.
* Explain IR detectors and the working of an IR camera.
* Write notes on IR imaging in aerospace and electronic components with a case study.

UNIT - V

Industrial Applications of NDE

1. Span of NDE Activities

* NDE is used in **every stage of the life of a product** :

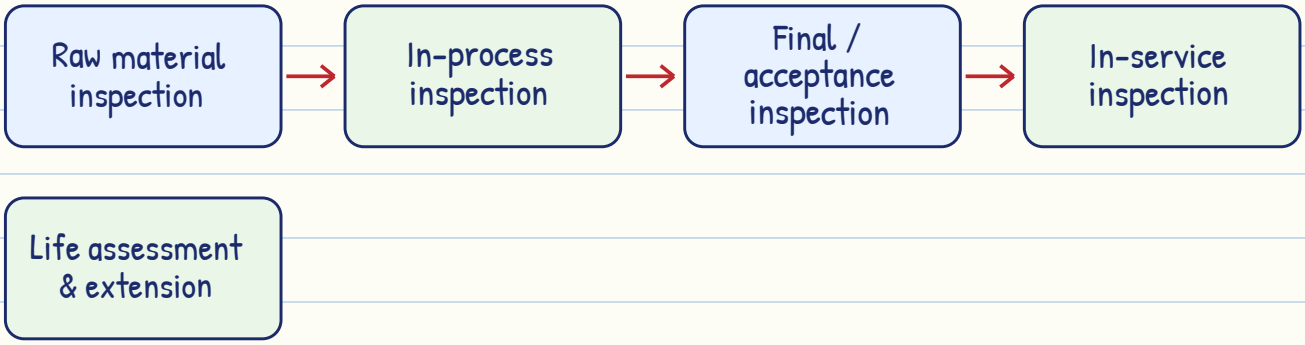


Fig: NDE through the life cycle

* **Raw material stage:** check ingots, plates, bars, forgings and castings for laminations, inclusions, cracks.
* **Manufacturing stage:** check welds, machined parts, heat treatment, assembly – feedback for process control.
* **Pre-service inspection (PSI)** – the baseline record before the plant starts.
* **In-service inspection (ISI)** – periodic inspection, to find fatigue cracks, corrosion, creep, erosion, stress corrosion cracking.
* **Condition monitoring and remaining life assessment.**
* **Failure analysis** after accidents.

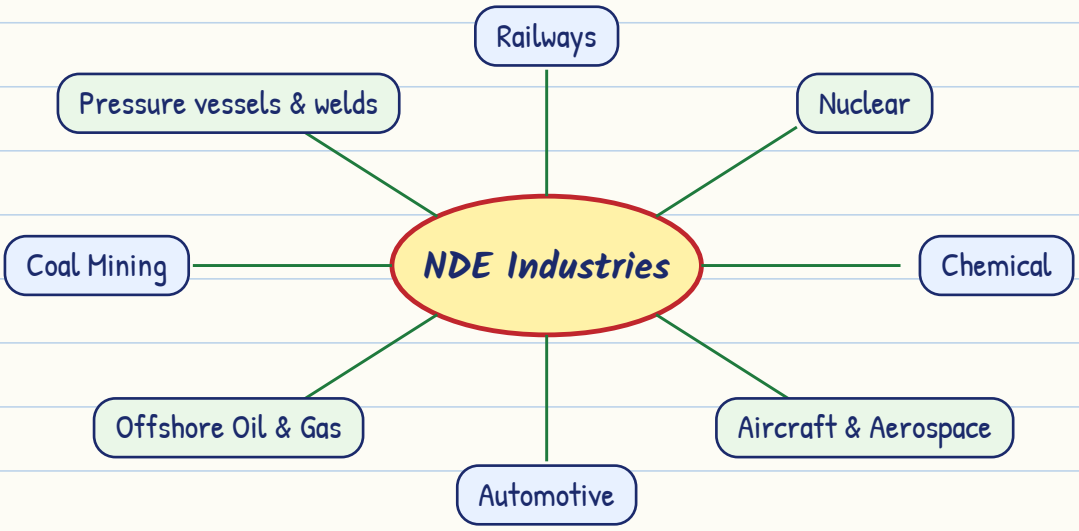


Fig: Span of NDE applications

Choice of NDE Method

Need	Preferred method
Surface cracks in ferromagnetic steel	MPT (with LPT as alternate)
Surface cracks in non-magnetic / non-porous	LPT, ET
Internal flaws in welds	RT, UT (TOFD, phased array)
Thickness and corrosion mapping	UT thickness gauge, RT profile
Tube inspection (heat exchanger)	Eddy current, remote field
Composite delamination, honeycomb	UT C-scan, thermography, shearography
Leak detection	Helium / bubble / acoustic / pressure decay
Hot spots, insulation, electrical	IR thermography
Fatigue crack growth monitoring	Acoustic emission

2. Railways

* Railways carry heavy loads at high speeds, so ***failure of rail, wheel or axle can cause derailment*** .
* ***Rail inspection: ultrasonic rail flaw detection*** using trains with probes at 0, 37 and 70 degree angles to find head checks, transverse defects (kidney flaws), ***detail fractures*** , web cracks and weld defects. Hand-pushed trolleys also used.

* **Eddy current testing** of rail head for **surface fatigue cracks (squats, gauge corner cracks)** and **magnetic flux leakage** for rails.
* **Rail weld inspection:** thermite / flash butt welds - UT and magnetic particle testing; radiography in some cases.
* **Wheels and axles: UT of axle (hollow and solid, from end face with angle probes)** to find fatigue cracks at press-fit seats; MPT after overhaul; **wheel tread ECT / UT** for cracks and shatter rim; wheel flat detection by **acoustic** or laser systems.
* **Bogie frames, couplers, springs, bolsters and castings** - MPT and LPT in workshops.
* **Bridges and tracks:** visual, UT of welded steel girders, rebound hammer and UPV on concrete sleepers, ground penetrating radar for ballast.
* **Hot box detectors** use passive **infrared sensors** to measure axle-box temperatures of passing trains.
* **Examples of defects:** rolling contact fatigue, corrugation, wheel burn, shelling.

3. Nuclear Industry

* Needs **the highest reliability and safety**; failure may cause radioactive release - every component is inspected by **strict codes (ASME Sec. XI, RCC-MR, AERB)** both before and during service.
* **Components:** reactor pressure vessel, steam generator tubes, primary coolant piping, fuel cladding (zircaloy tube), fuel pins, pressure tubes (PHWR), calandria, nozzle welds, turbines.
* **Methods used:**
  - **UT** - automated remote immersion scanning of reactor pressure vessel welds and nozzle to shell joints, **phased array and TOFD** on piping.
  - **Eddy current** - steam generator tubes (bobbin coil, rotating probes, array probes) to find wall thinning, pitting, fretting, stress corrosion cracks; **remote field ECT**.
  - **Radiography** - welds of piping and vessels, including high energy sources; **neutron radiography** for fuel and explosive devices.

- LPT / MPT for surface of components, leak testing (helium mass spectrometer) for containment and pipes.
- Visual inspection with remote cameras and underwater robots , acoustic emission for leak and crack monitoring.
- Gamma scanning, X-ray CT for fuel pellets and pins , dimension gauging of pressure tubes (ultrasonic wall thickness, diameter, sag).

* Special problems: high radiation field, limited access - need remote handling manipulators, robotic crawlers, shielding and dose control (ALARA) .
* Materials issues: irradiation embrittlement, stress corrosion cracking, hydrogen embrittlement (hydride blisters in zircaloy), thermal fatigue.
* Non-nuclear industries also use the same methods, but with lower requirements.

4. Non-nuclear and Chemical Industries

4.1 Non-nuclear (Thermal power, steel, general engineering)

* Boiler and turbine: boiler tubes - UT thickness, EMAT, RT of welds, ECT for tube thinning , creep and fatigue cracking; MPT of turbine blades, rotors, discs and shafts ; UT of rotor bores and forged shaft .
* Power plant: steam pipes (creep damage by replica metallography), headers, valve bodies, generator retaining rings (ECT), thermography of electrical boards , transformers and insulators.
* Steel plants: UT of rolled plates and bars for laminations, ET and MPT on billets, rolls; infrared of furnace refractory .
* Foundry and forge: RT, UT and MPT of castings and forgings.
* Bridges, buildings and civil structures: rebound hammer, ultrasonic pulse velocity (UPV), ground penetrating radar, half-cell potential for rebar corrosion, core test, MT on steel bridge weld.
* Wind turbines: blade UT, thermography, acoustic emission, tower weld MPT.

4.2 Chemical, Fertilizer and Petrochemical Industries

* Hazardous fluids at high pressure and temperature and corrosive environment -> leaks and failure cause fire, explosions and toxic release .

* **Equipment:** reactors, columns, heat exchangers, storage tanks and spheres, pipelines, furnaces, boilers, pumps, valves.
* **Typical damage:** general and **pitting corrosion, corrosion under insulation (CUI), erosion, stress corrosion cracking (SCC), hydrogen induced cracking (HIC), creep, fatigue, thermal fatigue, fouling** .
* **Methods:**
  - **UT thickness gauging and corrosion mapping (automated scanners, B/C-scan)** for tank floors and pipes.
  - **Radiography (including profile and real-time)** for CUI and weld inspection of pipe lines.
  - **Eddy current** on heat exchanger and condenser tubes, **IRIS (Internal Rotary Inspection System)** and **remote field ET** for ferromagnetic tubes.
  - **Magnetic flux leakage (MFL)** for tank floor and pipeline; **Guided wave UT** (long range) to check long pipe runs and under supports without removal of insulation.
  - **Wet fluorescent MPT** for SCC, **LPT** for stainless steel and non-magnetic alloys.
  - **Infrared thermography** - furnace tube temperature, refractory lining, steam trap, insulation loss.
  - **Acoustic emission** - online monitoring of pressure vessels and spheres, leak detection of valves.
  - **Risk based inspection (RBI)** decides the schedule and method according to risk.
* Inspection is done during **shutdown / turnaround** or on-stream.

5. Aircraft and Aerospace Industries

* **Safety critical** with high loads, fatigue and weight limits; **fail-safe and damage tolerance design** depend on NDE; inspection by regulations (DGCA, FAA, EASA) at specified flight-hour or cycle intervals (A, B, C, D checks).
* **Structures:** skin, fuselage lap joints, wing spars and ribs, landing gear, engine mounts, fasteners and rivet holes, control surfaces, bulkheads, propeller blades, helicopter rotor blades.

* **Typical defects:** fatigue cracks at fastener holes, corrosion, stress corrosion cracking, disbonds and delaminations, impact damage in composites, fretting, water in honeycomb.
* **Methods:**
  - **Eddy current** (surface probes, sliding probe for bolt holes, low frequency for sub-surface cracks under layers, conductivity for heat damage) - most used in-service method on **aluminium skin and engine disks** .
  - **Ultrasonic:** thick sections, forgings, bond testers (resonance, mechanical impedance analysis), **C-scan of composites** , phased array on wing, **Lamb waves** .
  - **Radiography:** castings, honeycomb water ingress, inaccessible regions; **real-time and digital radiography, CT** for turbine blades and cores.
  - **LPT (fluorescent)** on engine components, **MPT** on steel landing gear and bolts.
  - **Thermography** - delamination and disbond in composite panels, honeycomb, **space shuttle tiles** ; **shearography** for composite bond.
  - **Acoustic emission** for pressure tests and structural health monitoring; **tap test** for quick checking of the bonded skin.
  - **Borescope / video endoscope** for the inside of the engine and tanks.
* **Aerospace launch vehicles:** motor cases, cryogenic tank welds (RT, UT, helium leak test), solid propellant bond lines, nozzles; satellite structures.
* **Structural health monitoring (SHM)** with embedded sensors (fibre Bragg grating, piezo patches) for **real time** monitoring.

6. Automotive Industries

* Mass production and cost sensitive; need **fast, automatic, in-line** NDE on safety critical parts.
* **Engine: crank shaft, connecting rod, cam shaft, piston pin, gears** - **MPT** (automatic wet fluorescent benches) to find grinding cracks, forging laps; **ECT** for hardness, case depth, heat treatment, sorting of mixed parts.
* **Castings:** cylinder block and head, wheels (aluminium) - **radiography (real-time / digital X-ray, CT)** for porosity and shrinkage; **UT** for brake discs; LPT for leaks

and cracks.

* **Welds:** body in white **spot weld quality** by ultrasonic method (phased array), **laser weld** monitoring by IR, tube and pipes by ECT in production.
* **Brake and tyre: shearography / holography** of tyre sidewall and retreads, UT for brake lining bond, **eddy current for wheel rims** .
* **Fuel tanks, pipes, airbag inflators, radiators, AC lines - leak testing** (helium, pressure decay).
* **Adhesive joints** and composite bodies - ultrasonic, thermography.
* **Electronics and sensors** - X-ray for solder joints (BGA), IR thermography for hot spots in ECUs and battery cells (EV).
* **Final inspection:** vibration / acoustic noise tests (NVH), **eddy current sorting** of heat treated components.
* **Benefits:** zero defect supply, reduced warranty claims, recall prevention.

7. Offshore Gas and Petroleum Projects

* **Harsh environment:** seawater, waves and storms, fatigue, corrosion, marine growth, **remote access** and hazardous area; high cost of failure and pollution.
* **Structures:** jackets and platforms (steel tubular joints), risers, subsea pipelines, FPSO, drilling rigs, wellheads, pressure vessels and separators, crane, mooring chains, flexible risers, storage tanks.
* **Fabrication stage: RT, UT, MPT, LPT of welds** , mostly **AUT (automated UT), TOFD, phased array** for pipeline girth welds as per API 1104 / DNV; **alloy verification (PMI)** .
* **In-service inspection: underwater inspection by divers or ROV (remotely operated vehicle):**
  - **Visual (CCTV) and marine growth cleaning; flooded member detection (gamma or UT);**
  - **Alternating current field measurement (ACFM)** to find and size fatigue cracks at **tubular joint welds** even through coatings, without cleaning;
  - **Underwater MPT and eddy current; UT thickness** and corrosion mapping on risers, **cathodic protection (CP) potential survey** .

* **Pipeline pigging:** *intelligent pigs (in-line inspection tool)* run inside the pipeline using *magnetic flux leakage (MFL)* or *ultrasonic* sensors for finding corrosion, dents, cracks and wall loss over hundreds of kilometres.
* **Tanks and spheres:** floor corrosion by MFL, *acoustic emission* ; *guided wave* for risers and insulated lines.
* **Drilling equipment:** drill pipe, drill collars, BOP, wellheads - EMI (electromagnetic inspection), UT, MPT.
* **Leak detection:** acoustic, gas detectors, optical gas imaging with IR cameras.
* Standards: API 510, 570, 653, 1104, DNV-OS, NORSOK, ASME B31.3 and B31.8.

8. Coal Mining Industry

* Heavy-duty machinery in *dusty, wet, explosive* (methane and coal dust) atmosphere; *safety of miners* depends on integrity of equipment.
* **Hoisting equipment:** *wire ropes of winders and haulage* - the most important - inspected by *magnetic flux leakage (MFL) and magnetic rope testers* to find broken wires (loss of metallic area, LMA) and corrosion; *winding drums, shafts, sheaves, suspension gear* - MPT, UT; *cage and skip hangers, chains, hooks* - MPT.
* **Mining machines:** draglines, shovels, dumpers, conveyors, shearers, continuous miner - *UT and MPT of axles, shafts, gear, pins, welds in booms and buckets, bearing housings* ; *ECT* of hydraulic components.
* *Dump truck frames* and *rock bolts* - UT, *guided wave on rock bolts / anchor* .
* **Roof and rock mass:** *acoustic emission, microseismic monitoring* and *ground penetrating radar* for roof falls, strata monitoring, detection of voids, abandoned workings and water bodies.
* **Conveyor belts:** steel cord belt - X-ray or magnetic induction to find broken cords, joint failure; IR thermography of idlers and drive motors (fire prevention).
* **Pressure equipment:** boilers, compressors, air receivers - RT, UT.
* *Explosion proof and intrinsically safe NDT instruments* are required underground.

* **Thermography** to detect hot spots in coal stockpile (spontaneous heating), fires in coal seams.
* **Challenges:** dirt, grease, wet surfaces, dust, hazardous locations, limited time.

9. NDE of Pressure Vessels

* **Pressure vessels** (boilers, reactors, storage spheres, LPG bullets, air receivers, columns) hold fluids at **pressure different from the atmosphere** ; failure may be **catastrophic (explosion)** .
* Codes: **ASME Sec. VIII Div. 1 / 2, Sec. I (boilers), Sec. V (NDE methods), IS 2825, PD 5500, EN 13445** .
* **Parts:** shell, dished ends (heads), nozzles and manway, flanges, skirt / saddle, internals, welds (longitudinal, circumferential, nozzle to shell).
* **During fabrication:**
  - **Plates and forgings:** UT (laminations) as per ASTM A435 / A578; **PMI** for alloy.
  - **Weld edges and joints: MPT / LPT** for root and back chipping, **RT (spot or 100%, depending on joint efficiency 0.7 / 0.85 / 1.0)** and UT; **TOFD / PAUT** in place of RT for thick walls.
  - **Nozzle welds, attachment welds:** MPT or LPT, UT.
  - **Heat treatment check:** hardness test, **post weld heat treatment (PWHT)** records.
  - **Final hydrotest / pneumatic test** with **acoustic emission** monitoring.
* **In-service inspection:**
  - **External and internal visual,** UT thickness survey of shell and heads for corrosion allowance and remaining life; **wet fluorescent MPT** of welds (SCC, fatigue); **LPT on stainless steel; RT / UT / TOFD** for new crack growth; **hydrogen blistering and HIC** by UT shear wave (angle probes); **creep** by replica metallography and hardness; **thermography** for refractory; **acoustic emission** for on-line monitoring.
* **Inspection intervals** by **Risk Based Inspection (RBI)** and fitness for service (API 579).

* *Typical flaws:* weld cracks, lack of fusion, slag, porosity, corrosion pits, laminations, blisters, bulging.

10. NDE of Castings

* *Casting* = metal poured in a mould; defects occur in solidification and cooling.
* *Defects: blowholes and gas porosity, shrinkage cavity and porosity, hot tears and cold cracks, cold shuts, misruns, sand / slag inclusion, segregation, core shift, mismatch, surface roughness.*
* *Special problems of NDT on castings: coarse grain, rough surface, complex shape, varying thickness, scattering of ultrasound* (cast iron, steel, stainless).
* *Methods:*
  - *Visual inspection* and dimensional check - first step.
  - *Radiography* - the *best for volumetric flaws* (shrinkage, gas porosity, inclusion, hot tears); *film, real-time and CT for aluminium and titanium casting* ; graded by *ASTM E446 / E186 / E280 reference radiographs* .
  - *UT* - low frequency (0.5 - 2 MHz), large crystal, dual crystal probe, *also graded for porosity by back wall attenuation* ; good for thick steel castings.
  - *MPT* (steel, iron castings) for surface and near surface cracks, hot tears; *LPT* (aluminium, brass, stainless) for surface-opening flaws; dye penetrant on machined surface.
  - *Eddy current* for sorting, hardness and nodularity (ductile iron) checks.
  - *Leak tests, pressure test* for valve bodies, pump casing; thermal and *acoustic (ring) tests* .
* *Acceptance by classes* (such as ASTM A609, A903, ASME Sec. III NB-2500, MSS SP-54 for valves).
* *Typical castings:* pump casing, valve body, turbine blade, engine block, wheel hubs, gear box.

11. NDE of Welded Constructions

* *Welding defects* come from poor technique, wrong parameters, bad edge preparation or materials. *Weld inspection plan:*

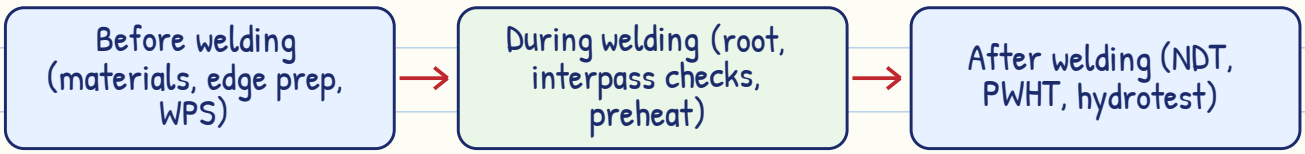


Fig: Three stages of weld inspection

* **Before welding:** check of base metal (UT for laminations), edge preparation, fit-up, WPS / PQR and welder qualification, **consumables** .
* **During welding:** visual of root pass, **MPT / LPT of root and interpass layers** , preheat and interpass temperature control.
* **After welding: visual testing first (VT)** , then **surface NDT** , then **volumetric NDT** , after cooling (hydrogen cracking may appear after 24 - 48 hours; for high strength steels wait before testing).

Typical weld discontinuities found by RT / UT / MT / PT

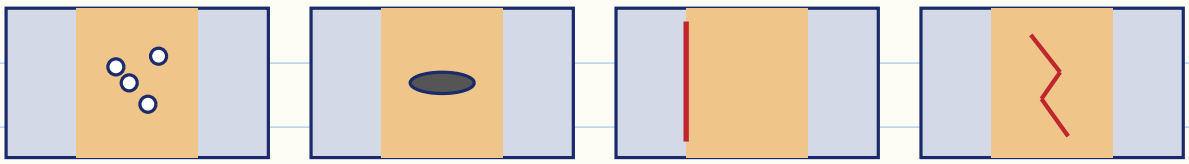


Fig: Common weld defects

Weld Defects and Suitable Methods

Defect	Best detection method
Porosity (round / cluster)	RT, UT (less sensitive)
Slag inclusion	RT, UT
Lack of fusion	UT, RT (if aligned), MPT for surface
Incomplete penetration	RT, UT
Cracks (hot, cold)	UT, MPT / LPT for surface, RT if favourable
Undercut, overlap, spatter	Visual, MPT / LPT
Tungsten inclusion	RT (bright spot)
Distortion, misalignment	Visual and dimensional gauges

Lamellar tearing	UT (normal probe), MPT

* **Radiography** - for butt welds; permanent record; planar flaws may be missed.
* **UT with angle beam, TOFD, phased array** - best for planar flaws in thick welds and in-service inspection; automated for pipelines.
* **MPT / LPT** - for surface and near surface cracks in ferromagnetic / non-magnetic welds.
* **Visual testing** - up to 80% of flaws found; uses weld gauges, magnifier, borescope.
* **ECT / ACFM** - for surface cracks at welds in tubes and offshore joints.
* **Leak testing and hydrotest** - proof of tightness and strength.
* **Acceptance criteria:** ASME Sec. VIII (UW-51 / UW-52), ASME B31.1 / B31.3, API 1104, AWS D1.1 (statically / cyclically loaded structures), ISO 5817 quality levels B / C / D, IS 4853, 2595, 3658, 7307.
* **Records:** NDT procedure, technique sheet, qualified personnel (**ISO 9712 / SNT-TC-1A Level I, II, III**), report with weld map.

Qualification of NDT Personnel

* **SNT-TC-1A, ISO 9712, NAS 410, ISNT (India)** - Level I (performs tests), Level II (sets up and calibrates, interprets, reports), Level III (develops procedures, examines, trains).
* Certification needs training, experience, vision test and examination.

12. Recent Trends in NDE

* **Digital radiography and computed tomography, phased array UT, TOFD, guided wave UT, EMAT, thermography, shearography, laser UT, terahertz imaging, acoustic emission with AI.**
* **Automation and robotics, drone inspection (tanks, chimneys), crawlers, data analytics, machine learning for defect recognition, digital twin and NDE 4.0.**
* **Structural health monitoring (SHM)** - sensors on the structure monitor continuously.
* **Integration with RBI and fitness for service (FFS)** - decisions based on the size of flaw and life.

Summary:

NDE is applied from raw material to in-service in every industry. Railway – UT / ET on rail, axle; Nuclear – remote UT / ECT, strictest; Chemical – corrosion mapping, tubes; Aerospace – ECT, UT, thermography for composites; Automotive – fast MPT, ET, RT in line; Offshore – ROV, ACFM, pigs; Mining – rope MFL; Pressure vessels, castings and welds – RT + UT + MPT / LPT.

Important Questions

* Explain the NDE methods used in the railway and nuclear industries.
* Discuss NDE in aircraft and aerospace industries including composite structures.
* Write NDE applications in the offshore gas, petroleum and coal mining industries.
* How are pressure vessels inspected during fabrication and in service?
* List the defects in castings and welds and the suitable NDT methods for each.